

Supporting Information

Crystal Facet Engineering on SrTiO₃ Enhances Photocatalytic Overall Water Splitting

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Experimental Section

Materials. Titanium chloride solution (15.0~20.0 % TiCl_3 in 30 % HCl , AR) was purchased from Aladdin. Strontium chloride hexahydrate ($\text{SrCl}_2 \cdot 6\text{H}_2\text{O}$, 99.99%) Aluminum chloride hexahydrate ($\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$, AR), and ethanol (EtOH, AR) were bought from Sinopharm Chemical Reagent. Lithium hydroxide monohydrate ($\text{LiOH} \cdot \text{H}_2\text{O}$, 98%) and ethylene glycol were received from Alfa Aesar.

All the chemicals were used as received without any further purification. Ultrapure water (Millipore Milli-Q grade) with a resistivity of $18.2 \text{ M}\Omega \text{ cm}$ at 25°C was used throughout all the experiments.

Synthesis of nanosphere SrTiO_3 (STO) precursor. The nanosphere STO precursor was synthesized through a modified hydrothermal reaction, according to a previously reported method.¹ Typically, 0.8 mL of TiCl_3 solution was added into 25 mL ultrapure water containing 0.9 mL ethylene glycol cooled in an ice bath and stirred for 5 min. 30 mL of LiOH solution containing 4.0 g $\text{LiOH} \cdot \text{H}_2\text{O}$ was then added. After stirring for 15 min, 0.74 g $\text{SrCl}_2 \cdot 6\text{H}_2\text{O}$ was added and stirred for additional 10 min. Subsequently, the mixed feedstock was transferred to a 100 mL Teflon-lined stainless-steel autoclave and reacted at 180°C for 1~48 h. After reaction, the autoclave was submerged directly in ice water for rapid cooling and the precipitate was recovered by centrifugation, washed with ultrapure water and ethanol until no precipitation in the separated supernatant when the AgNO_3 solution was added and then dried at 60°C in vacuum drying chamber.

Synthesis of cubic mesoporous STO precursor. The hydrothermal reaction procedure is consistent with that of nanosphere STO precursor, and the reaction was maintained at 180°C for 48 h. After the reaction, the autoclave was cooled naturally and the precipitate was recovered by centrifugation, washed with ultrapure water and ethanol until no precipitation in the separated supernatant when the AgNO_3 solution was added and then dried at 60°C in vacuum drying chamber.

Synthesis of cuboctahedron STO. The cuboctahedron STO was synthesized in $\text{SrCl}_2 \cdot 6\text{H}_2\text{O}$ molten salt.² In a typical synthesis, the prepared nanosphere STO precursors and $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$ were mixed and dispersed in an appropriate amount of ultrapure water in a molar ratio of 1:0.004 (Donated as 0.4 mol%, where x mol% represents the Al/Ti molar ratio in the starting mixture), and evaporated in an evaporation dish with constant stirring to complete dryness. The resultant mixture was collected and mixed with $\text{SrCl}_2 \cdot 6\text{H}_2\text{O}$ in a molar ratio of 1:10 and then heated at 1123~1523 K for 10 h in air using an alumina crucible. The heating and cooling programs are set at 4 K/min and 2.5 K/min, respectively. After cooling to room temperature, the product was separated from excess SrCl_2 by centrifugal washes with ultrapure water for three times and then dried at 60°C in vacuum drying chamber.

Synthesis of cubic STO from cubic mesoporous STO precursor. Except for the utilizing of cubic mesoporous as precursor, the rest of the synthesis procedure is

consistent with cuboctahedron STO.

Characterization of materials. The crystal structure of as-prepared samples was determined by X-ray diffraction (XRD) from Bruker D8 Advanced Diffractometer with Cu K α radiation ($\lambda = 1.5406 \text{ \AA}$) at the scanning speed of $4^\circ/\text{min}$. The morphology was examined by field emission scanning electron microscope (FE-SEM, HITACHI S4800). Raman spectra were recorded on a Renishaw In Via Raman microscope in the range of $100\text{--}1200 \text{ cm}^{-1}$ under a 514 nm laser excitation. The light absorption range of the photocatalysts were carried out by ultraviolet-visible (UV-vis) spectrophotometry with a UV-vis spectrophotometer (CARY 500). X-ray photoelectron spectroscopy (XPS) was performed on ESCALAB 250 Xi X-ray photoelectron spectrometer using a monochromatized Al K α X-ray source ($h\nu = 1486.6 \text{ eV}$) to investigate surface elements content and valence of as-prepared samples. The XPS data were calibrated by the binding energy of C 1s peak at 284.8 eV. X-band electron paramagnetic resonance (EPR) spectra was performed with a Bruker ER200D instrument at 9.86 GHz and 20 mW to analyze electronic structural changes. Ultraviolet photoelectron spectroscopy (UPS, PHI 5000 Versaprobe II) was conducted to determine the VBM of semiconductors using He I (21.22 eV) as the excitation source at an applied bias voltage of 10 eV. Time-resolved photoluminescence (TR-PL) spectra was conducted on Fluorolog-3 (HORIBA) to investigate the precise life time of electron-hole pairs.

Aberration-corrected STEM characterization was performed on a ThermoFisher Themis Z microscope equipped with two aberration correctors under 300 kV. High angle annular dark field (HAADF)-STEM images were recorded using a convergence semi angle of 17.8 mrad, and inner- and outer collection angles of 47 and 200 mrad, respectively. Energy dispersive X-ray spectroscopy (EDS) was carried out using 4 in-column Super-X detectors. Integrated differential phase contrast (iDPC)-STEM was recorded using a camera length of 285 mm on DF4 detectors, with inner- and outer collection angles of 5 and 26 mrad, respectively.

Ti and Sr K-edge XAFS spectra were performed at BL14W1 beamline in Shanghai Synchrotron Radiation Facility (SSRF), operated at 3.5 GeV with injection currents of 140-210 mA. The Ti K-edge XAFS measurements were performed in fluorescence mode. Si {111} double-crystal monochromators were used to reduce the harmonic component of the monochrome beam. The acquired EXAFS data were processed according to the standard procedures using the ATHENA module implemented in the IFEFFIT software packages. First, subtracting the post-edge background from the overall absorption and normalizing with respect to the edge jump step to obtain EXAFS spectra is necessary. Then, $\chi(k)$ data in the k-space (ranging from 3 to $12 \text{ \AA}^{-1}$ for Ti and Sr) were Fourier transformed to real (R) space using hanning windows

Loading of Rh/Cr₂O₃/CoOOH cocatalysts. Aqueous solutions of RhCl₃·6H₂O, K₂CrO₄ and Co(NO₃)₂·6H₂O with Rh, Cr, Co concentration of 2mg/mL were prepared. In a typical photodeposition process, 100 mg of catalyst was dispersed in 100 mL of ultrapure water by 10 min sonication and then appropriate amount of the RhCl₃ and Co(NO₃)₂ aqueous solution were simultaneously added into the reaction suspension

with magnetic stirring. The suspension was irradiated with a Xe lamp (300 W, full arc) for 60 min, and then a certain amount of K₂CrO₄ aqueous solution was added and irradiated for another 30 min. The *in situ* photo-deposition serves to demonstrate the anisotropic separation of carries between {100} and {111} facets. The increased loading mass of Rh (0.4 wt%)/Cr₂O₃ (0.2 wt%)/CoOOH (0.4 wt%) facilitates clearer observation of the deposition position using SEM/TEM, and the loading amount doesn't affect where it's positioned in the crystal structure.

Photocatalytic overall water splitting performance evaluation. The photocatalytic water splitting reactions were carried out in a Pyrex top-irradiation reaction vessel connected to a glass gas-closed-circulation system (Perfectlight Sci & Tech, Labsolar 6A) with a top irradiation-type reaction vessel. The reaction solution was evacuated several times to remove air completely prior to irradiation under a 300W xenon lamp (>300nm, Perfectlight Sci & Tech, PLS-SXE-300+). The temperature of the reaction solution was maintained at 298K by the flow of cooling water during the reaction. The evolved gases were analysed by gas chromatography (Fuli, GC-9790-II), with a thermal conductive detector and 5Å molecular sieve column, using Ar as the carrier gas. Typically, 100mg catalysts loaded with Rh/Cr₂O₃/CoOOH were dispersed in 100 mL of ultrapure water with magnetic stirring and the evolved gas was detected every hour under continuous illumination.

Apparent quantum yield measurements. The H₂ evolution rate was measured in the same reaction system as the other photocatalytic reactions. To determine the wavelength dependence of the AQY, samples were irradiated with monochromatic light generated by an Xe lamp and sent through bandpass filters with central wavelengths of 350 nm, 360 nm, 365 nm, 370 nm or 380 nm (Optics). The full-width at half-maximum of each of these bandpass filters was approximately 10 nm.

The AQY was calculated according to the following equation:

$$AQY = \frac{\text{Number of reacted electrons}}{\text{Number of incident photons}} \times 100\% = \frac{2 \times n \times N_A}{S \times P \times t \times \frac{\lambda}{h \times c}} \times 100$$

where n is the amount of H₂ molecules (μmol) per hour; N_A is the Avogadro constant (6.022 × 10²³ mol⁻¹); S is the irradiation area (cm²); P is the monochromatic light intensity (mW/cm²); t is the light irradiation time (s); λ is the wavelength of the monochromatic light (nm); h is the Plank constant (6.626 × 10⁻³⁴ J s); c is the speed of light (3 × 10⁸ m·s⁻¹).

Taking monochromatic light at 350nm measurement (Tab. S1) as an example, AQY was calculated as follows:

$$AQY = \frac{\text{Number of reacted electrons}}{\text{Number of incident photons}} \times 100\% = \frac{2 \times n \times N_A}{S \times P \times t \times \frac{\lambda}{h \times c}} \times 100\%$$

$$= \frac{2 \times 384.1 \times 10^{-6} \times 6.022 \times 10^{23}}{24.6 \times 9.74 \times 10^{-3} \times 3600 \times \frac{350 \times 10^{-9}}{6.626 \times 10^{-34} \times 3 \times 10^8}} \times 100\%$$

$$= 30.5\%$$

Table S1 AQY calculation. Calculated AQY values of CoOOH/Rh/Cr₂O₃-modified Cubo-STO (Rh, Cr, Co loading amount on SrTiO₃, 0.2 wt%, 0.1 wt%, 0.2 wt%) for photocatalytic overall water splitting.

Wavelengths (λ, nm)	H ₂ evolved (μmol/h)	Catalysts mass (mg)	Light power (mW/cm ²)	AQY(%)
280	114.3	100	3.15	35.0
313	118.5	100	2.85	35.9
335	252.3	100	5.80	35.2
350	384.1	100	9.74	30.5
360	275.7	100	8.31	24.9
365	215.6	100	8.50	18.8
370	195.1	100	8.65	16.5

Solar-to-hydrogen energy conversion efficiency measurements. The STH efficiency was measured under simulated sunlight irradiation (AM 1.5G, 24.6 cm² illuminated area.). The STH efficiency was determined according to the following equation:

$$STH(\%) = \frac{R(H_2) \times \Delta G_r}{P \times S} \times 100$$

where R(H₂), ΔG_r, P and S represent the H₂ evolution rate, the reaction Gibbs energy of water splitting, the energy intensity of the AM 1.5 G solar irradiation (100 mW cm⁻²) and the irradiated sample area, respectively.

DFT calculations. The Vienna Ab-initio Simulation Package (VASP) code³ within the projector augmented wave (PAW) formalism⁴ is used in the present density functional theory (DFT) calculations. The generalized gradient approximation (GGA) for the exchange correlation functional within the Perdew-Burke-Ernzerhof (PBE)⁵ parametrization is employed. The cutoff energy for the wave-function expansion is 520 eV to ensure robust energy convergence.

To determine the band-edge energy alignment between {100}, {110}, and {111} facets with different atomic terminations, we construct supercell slabs along the [100], [110], and [111] orientations of the SrTiO₃ crystal, incorporating various terminations. The thickness of the vacuum slab (about 30 Å) is approximately equal to the thickness of the bulk slab.

The {100} slabs are analyzed using a k-point grid density of 6 × 6 × 1 in the first Brillouin zone. For the {110} slabs, a 6 × 4 × 1 k-point grid is utilized, and for the {111} slabs, a 4 × 4 × 1 k-point grid is employed. During structural relaxations, all atoms are

permitted to relax until the Hellmann-Feynman forces acting on them are smaller than 0.02 eV/Å.

Figure S2 shows the average electrostatic potential between the bulk-like region and vacuum-like region in the supercells with various atomic terminations. Due to the discrepancy between different terminations and directions of these typical facets, although the band gap of SrTiO₃ remains unchanged, their band-edge energy levels [valence-band maximum (VBM) and conduction-band minimum (CBM)] are no longer identical.

To calculate the surface energy of these typical facets with different terminations more reasonably, we take the chemical potential of each element into account. The surface energy is defined as:

$$E_{surface} = \frac{E_{relaxed} - \sum_i [n_i(\mu_i + E_i)]}{2S}$$

where n_i , μ_i , and E_i are the number of atoms, the chemical potential, and the total energy per atom of elementary substance of a specific element in the supercell (here subscript i could be Sr, Ti, or O), respectively. $E_{relaxed}$ is the total energy of the relaxed superlattice. S represents the surface area.

A thermodynamically consistent process for determining the experimentally relevant range of chemical potentials is outlined below. The total energy of a stoichiometric unit of SrTiO₃ can be expressed as:

$$\mu_{Sr} + \mu_{Ti} + 3\mu_O = E_{SrTiO_3}^{bulk} - E_{Sr}^{bulk} - E_{Ti}^{bulk} - 3E_O^{gas} = \Delta H_{SrTiO_3}$$

where ΔH_{SrTiO_3} is the formation enthalpy of SrTiO₃.

The stability of the system against decomposition into its constituent elements places upper limits on the chemical potential of each element in SrTiO₃. The reference substances are pure solids of Sr and Ti, as well as molecular oxygen (O₂). Therefore, additional restrictions are determined by:

$$\mu_{Sr} \leq 0$$

$$\mu_{Ti} \leq 0$$

$$\mu_O \leq 0$$

Since the binary oxides like SrO, SrO₂, TiO, and TiO₂ do not precipitate from bulk

SrTiO₃, the following restrictions are also included:

$$\mu_{\text{Sr}} + \mu_{\text{O}} \leq \Delta H_{\text{SrO}}$$

$$\mu_{\text{Sr}} + 2\mu_{\text{O}} \leq \Delta H_{\text{SrO}_2}$$

$$\mu_{\text{Ti}} + \mu_{\text{O}} \leq \Delta H_{\text{TiO}}$$

$$\mu_{\text{Ti}} + 2\mu_{\text{O}} \leq \Delta H_{\text{TiO}_2}$$

where ΔH_{SrO} , ΔH_{SrO_2} , ΔH_{TiO} , and ΔH_{TiO_2} are the formation enthalpies of SrO, SrO₂, TiO, and TiO₂.

The above set of equations defines the range of chemical potentials consistent with the stability of SrTiO₃ against decomposition into binary oxides or into its elemental components. Given the boundary for the chemical potential of each element, a valid range for the surface energies of the typical facets with various termination can be examined.

After comprehensively considering the abovementioned chemical potential restrictions, a well-defined ternary phase diagram of SrTiO₃ is determined and shown in **Figure S2**. This figure depicts a narrow region of stability for SrTiO₃, defined by the quadrilateral-like area filled in green, and restricted by the Sr, Ti, O₂, SrO, SrO₂, TiO, and TiO₂ precipitation lines.

The two corner points, A and B, correspond to O-poor and O-rich precipitate phase boundaries, respectively. The middle point C between A and B denotes the O-medium condition. For the {100} slabs, to distinguish the stability between different terminations, a yellow ribboned area is marked in the green region. In the yellow area, TiO₂ termination exhibits greater stability, whereas in the left green region, SrO termination is more stable.

In experiment, the termination of the {100} facet is TiO₂. Moreover, since the highly-crystalline and well-defined STO single crystal was prepared in SrCl₂ melt salt, it is reasonable to set the chemical potential restriction as the Sr-rich condition. To be consistent with the experimental conditions, the corner point D in the yellow ribboned area is chosen to be the case that is closest to experiment. The specific values of the chemical potentials for Sr, Ti, and O at the four points (A, B, C, and D) mentioned above are given in **Table S2**. Based on the value at point D, the surface energies of several different crystal facets in the manuscript can be evaluated.

Table S2 Chemical potentials of Sr, Ti, and O at points A, B, C, and D in Figure S1.

	$-\mu_{\text{Sr}}$	$-\mu_{\text{Ti}}$	$-\mu_{\text{O}}$
A: O-poor	0.37	0.01	5.13

B: O-rich	6.6	9.17	0
C: O-medium	3.49	4.6	2.56
D: Sr-rich	2.56	1.11	4.03

Simulated photocarrier distribution in light-illuminated cuboctahedron STO. The photocarrier distribution in Cubo-STO with {100} and {111} facets was simulated by the semiconductor module of COMSOL Multiphysics software. A two-dimensional hexagon model was used to represent the cross section of Cubo-STO particles, whose geometry was deduced from FE-SEM images. The semiconductor parameters of STO were collected from the literature and are tabulated in **Table S3**. The simulation was performed based on the finite volume method that used Scharfetter–Gummel discretization for the calculation of particle transport. To simplify the calculation, photocarriers were assumed to be homogeneously generated within Cubo-STO particles since random particle stacking and multiple photon reflection/scattering average photon absorption. Defect-assisted photocarrier recombination (i.e., Shockley-Read-Hall type) was considered. The interfaces at all crystal facets were assumed to be Schottky-type ones to represent the semiconductor–electrolyte interfaces. The model was discretized with a finemesh before calculation. The calculation was performed by self-consistently solving Poisson, drift-diffusion, and current continuity equations at all discretized nodes to reach convergence.

Table S3 Parameters used in the simulation for electron/hole densities.⁶

Parameters	SrTiO ₃
Conduction band effective density of states	$7.94 \times 10^{20} \text{ cm}^{-3}$
Valence band effective density of states	$6.71 \times 10^{20} \text{ cm}^{-3}$
Charge carrier mobility	$6 \text{ cm}^2 \text{ V}^{-1} \text{ S}^{-1}$
Band gap	3.2 eV
Electron affinity	4.2 eV
Dielectric constant	300
Photocarrier generation rate	1×10^{16}
Charge carrier lifetime	24 ns

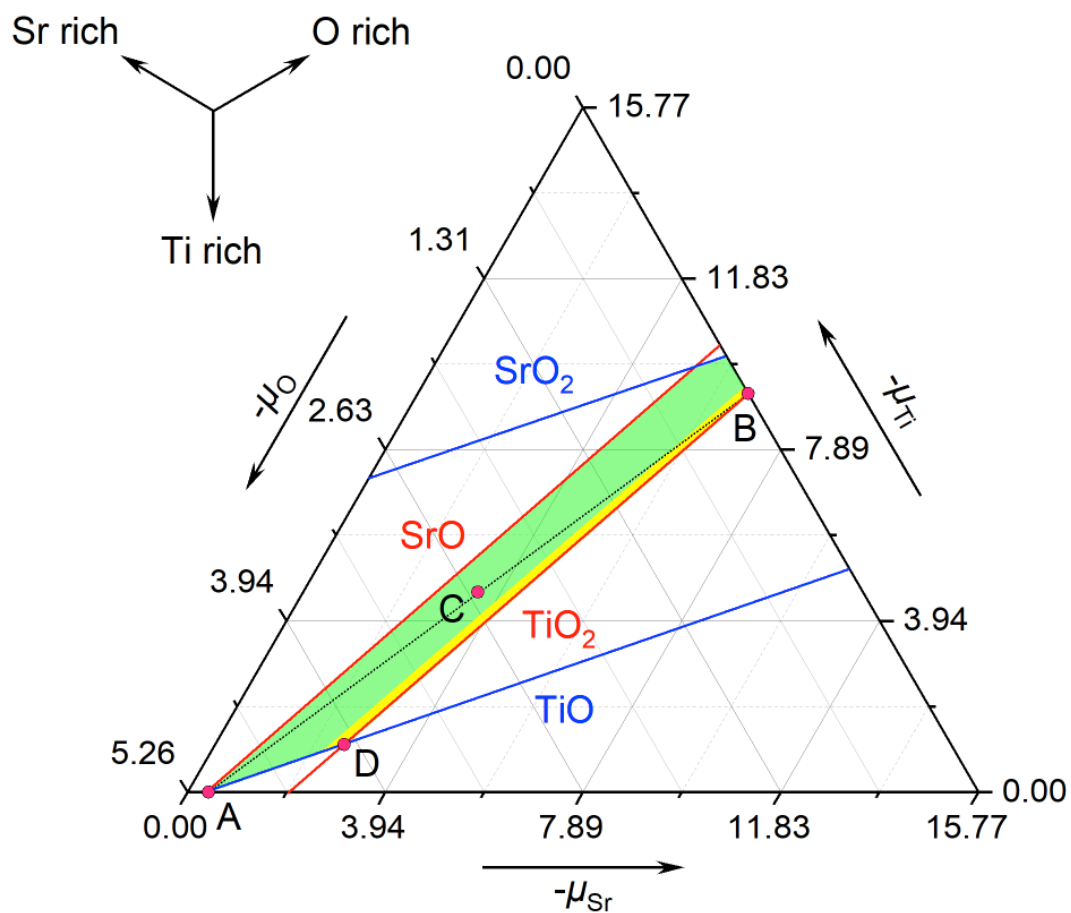


Figure S1 Calculated ternary phase diagram. SrTiO₃ is stable in both the green and yellow areas. For the {100} slabs, TiO₂ termination exhibits greater stability in the yellow region, whereas SrO termination is more stable in the green region.

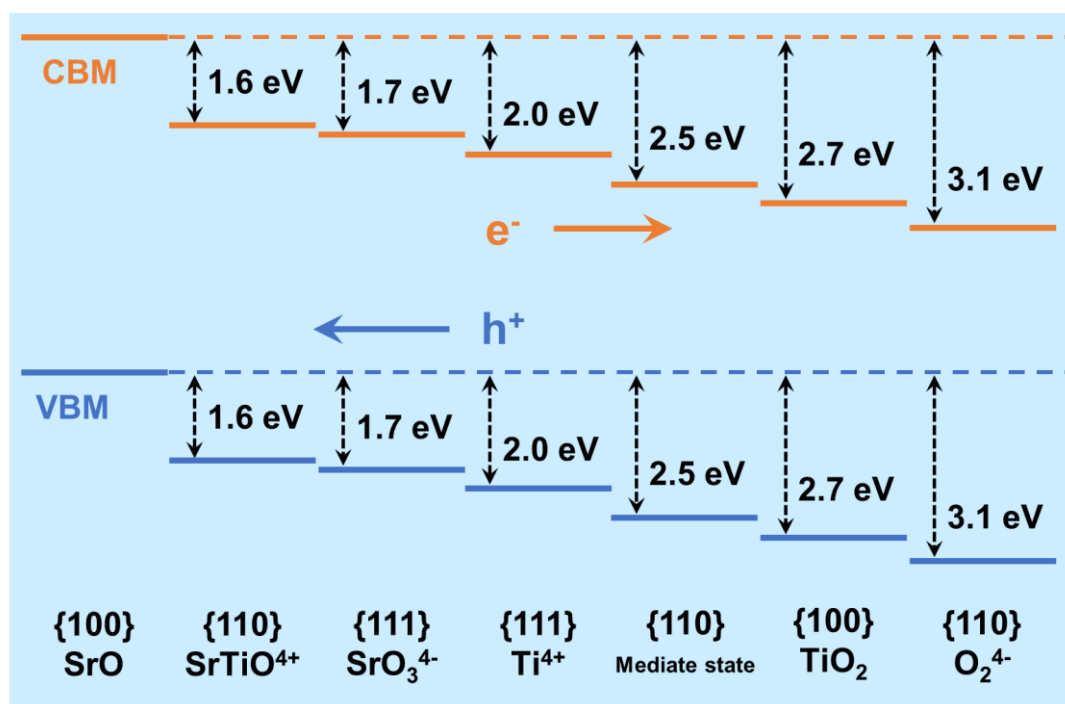


Figure S2 Calculated band alignment. Calculated energy of {100}, {110} and {111} facets terminated with various atomic termination.

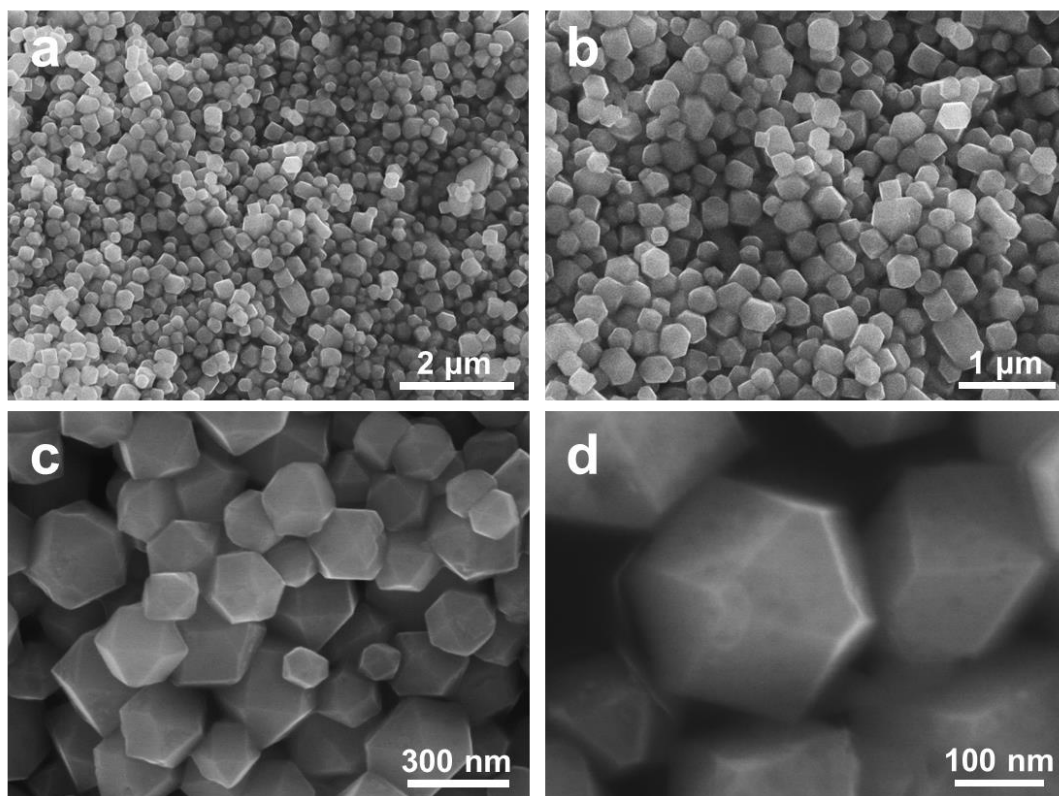


Figure S3 SEM images of Cubo-STO single crystals. a-d) SEM images of typical Cubo-STO at different amplification, revealing the uniformity of single-crystal morphology.

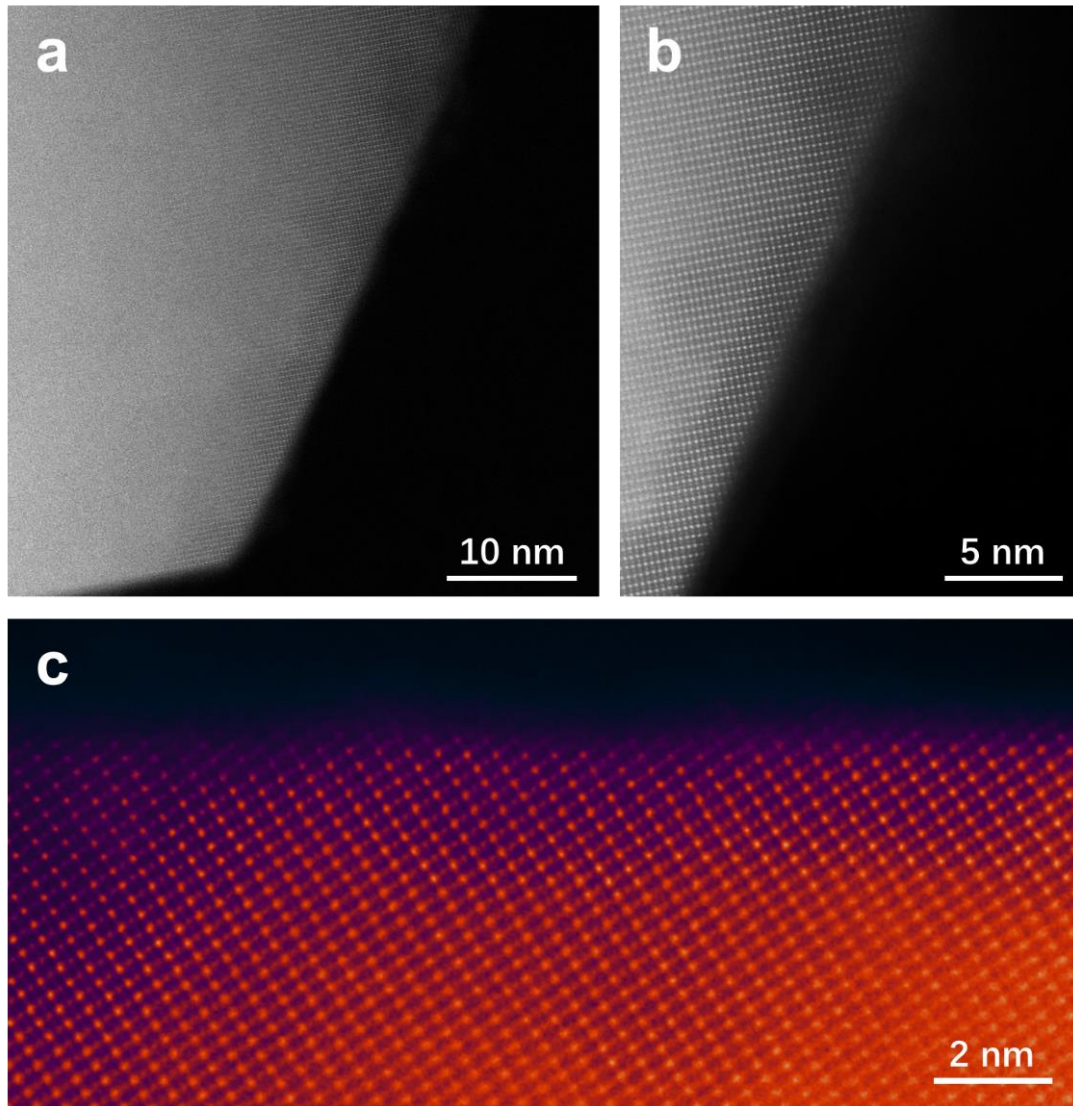


Figure S4 HAADF-STEM images of Cubo-STO single crystal along the $[1-10]$ zone axis. a, b) HAADF-STEM images showing smooth $\{111\}$ surface at different magnifications. c) Atomic-scale HAADF-STEM image after directional correction, which was a supplement to $\{111\}$ facet that not fully shown in **Figure 2c** at the article.

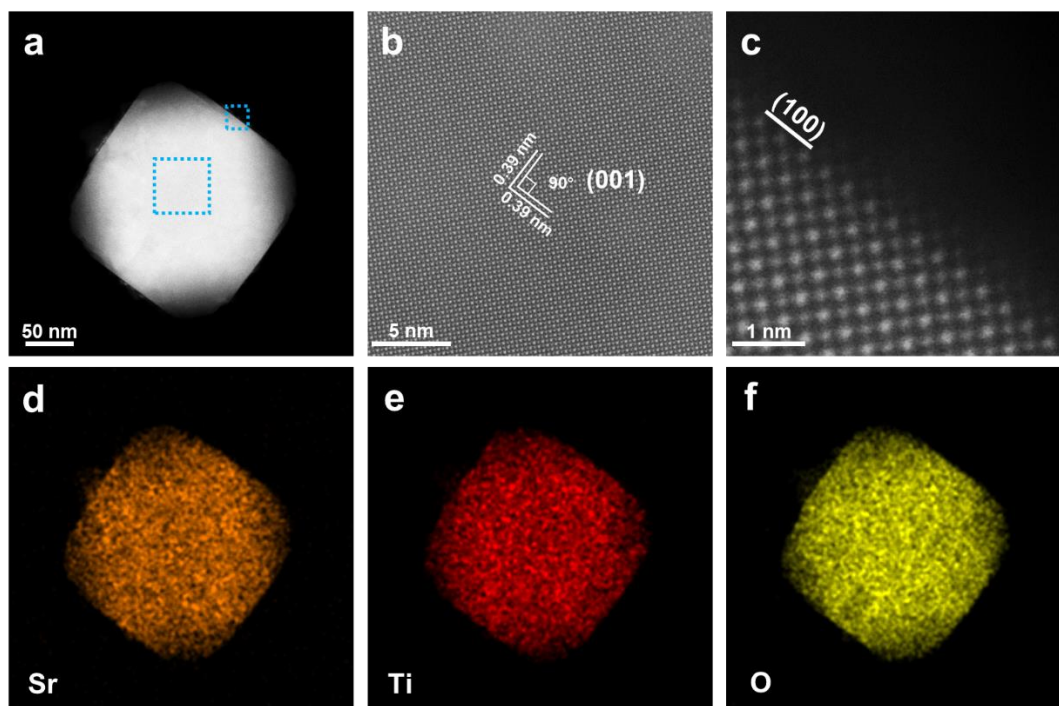


Figure S5 STEM characterization of Cubo-STO single crystals. a) HAADF-STEM image of cuboctahedron STO along the $[001]$ zone axis. b, c) Atomic-scale images of the two blue dashed-line region in (a), respectively, displaying smooth and clean surface. d-f) The EDS maps of Sr, Ti, O elements from the crystal in (a). Note: Strontium titanate is a cubic phase structure with six strictly symmetrical and identical square crystal faces, all referred as $\{100\}$ facet.

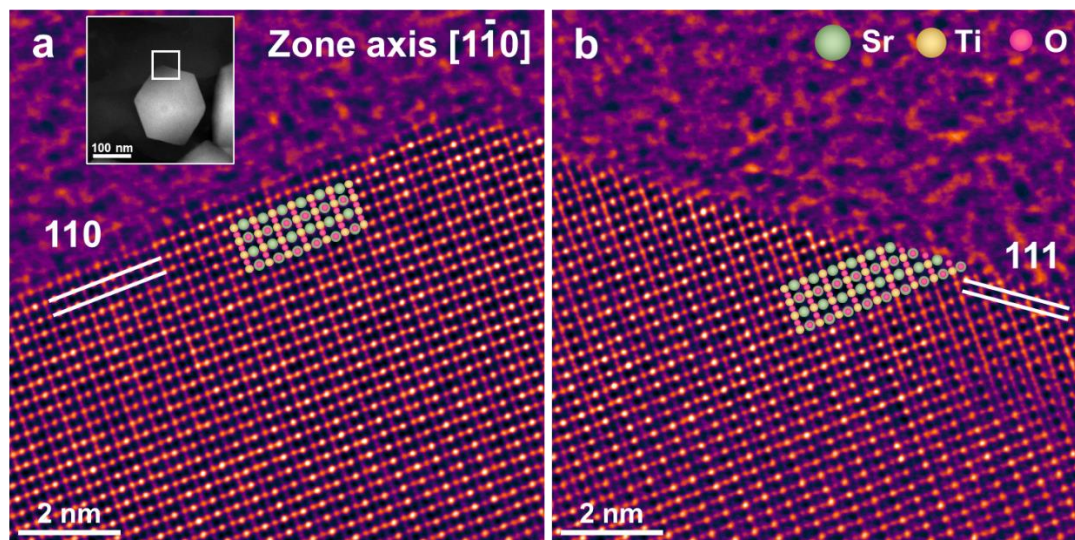


Figure S6 False-colored iDPC-STEM images of part of typical Cubo-STO single crystal along the $[1\bar{1}0]$ zone axis. The atomic-scale IDPC-STEM images showing the saved edges along $[110]$ orientation (a) and the flat $\{111\}$ surface (b) taken from the region in the inset. Note: at the vertex of the cuboctahedron where two $\{111\}$ facets contact, several nanometer-scale edges were saved along the $[110]$ orientation and the terminating atomic configurations are irregular coexisting with SrTiO^{4+} and an intermediate state (O_2^{4+} with partial oxygen loss).

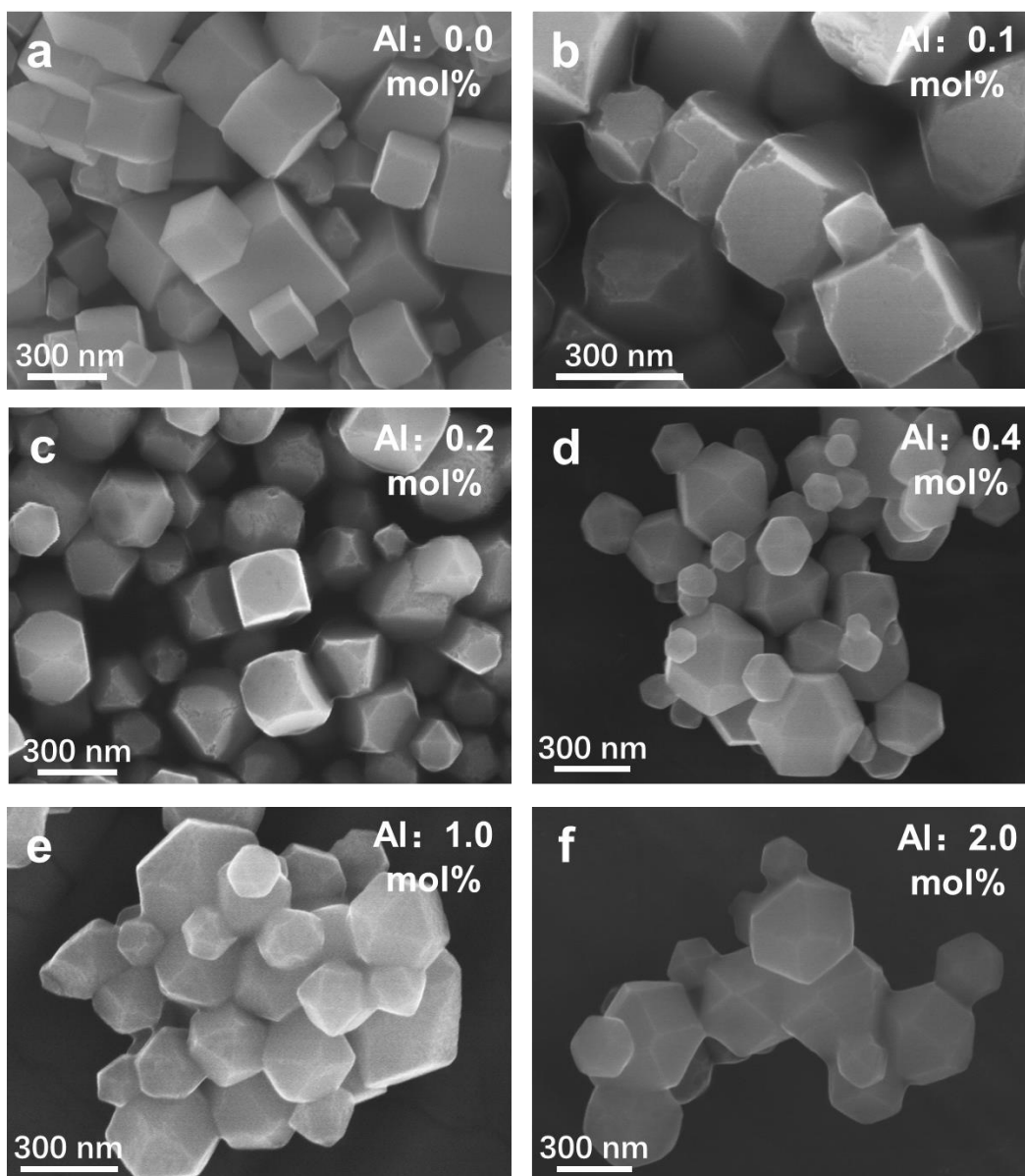


Figure S7 Effect of Al doping on crystal morphology. a-f) SEM images of a series of STO crystal particles with various amounts of Al doping, indicating a serial evolving process from cube to cuboctahedron morphology and changeless morphology with amount of Al doping above 0.4 mol%.

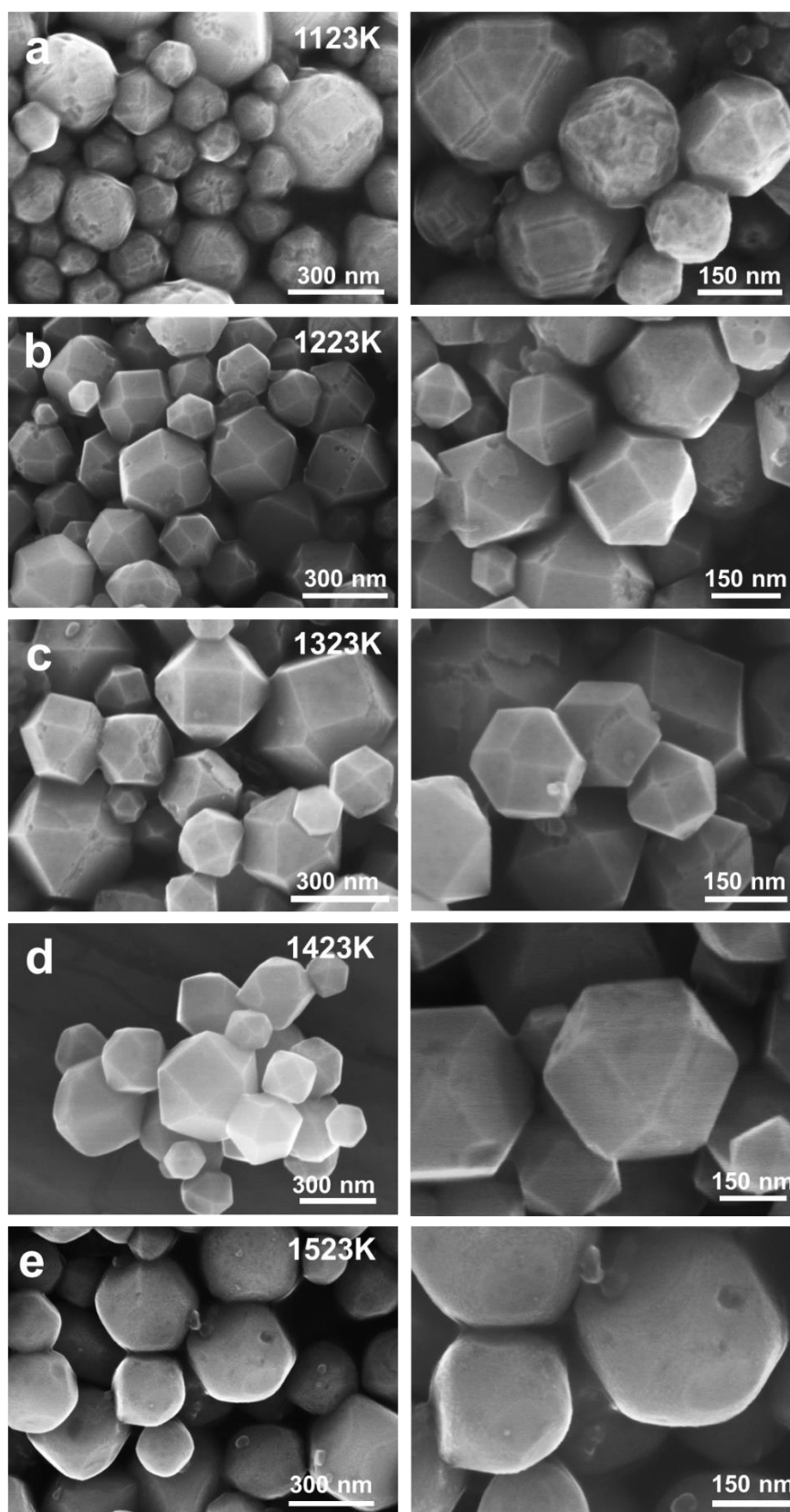


Figure S8 SEM images of typical Cubo-STO prepared at different temperatures. SEM images of samples calcined with the temperature from 1123K to 1523K, the well-defined cuboctahedron morphology was observed in the temperature range of 1223K to 1423K.

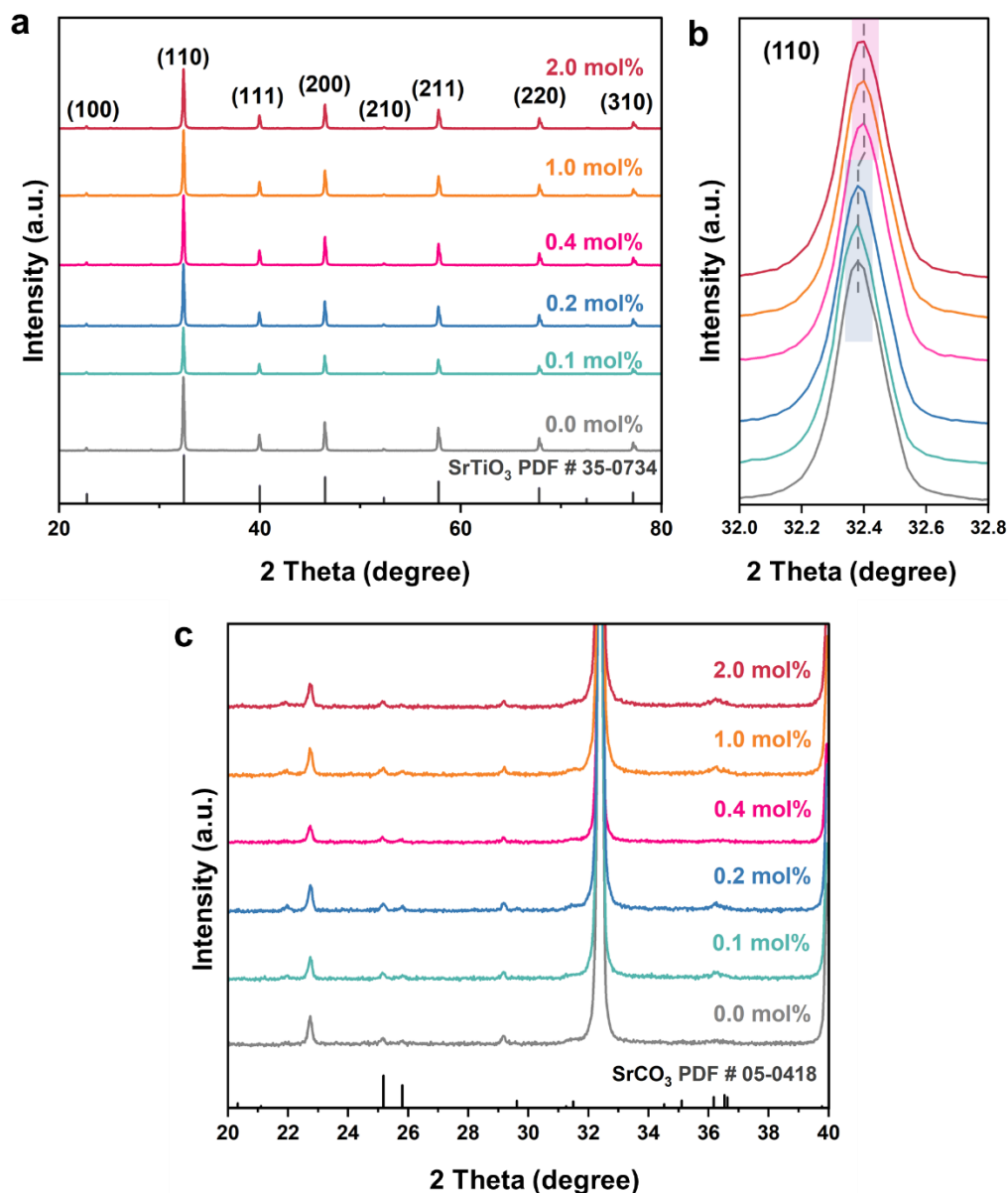


Figure S9 Effect of Al^{3+} doping on crystal structure. The XRD patterns of a) as-prepared STO crystals with various amounts of Al additive, b) the enlarged {110} diffraction peak region, and c) the enlarged region ranging from 20 to 40 degrees in (a)

A constant lattice shrinkage was present for the samples with Al^{3+} additive above 0.4%, indicating the doping in bulk occurs drastically only with additions up to 0.4% and the few Al^{3+} below 0.4% work mainly on the surface to control morphology.

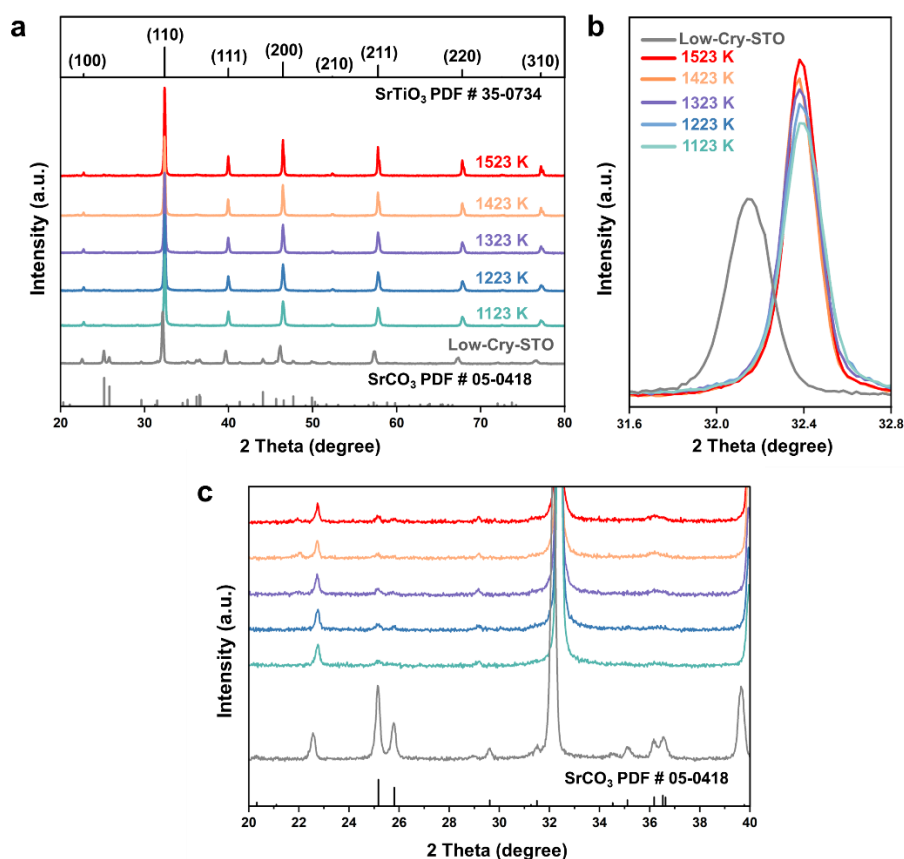


Figure S10 The crystal structure of typical Low Crystalline nanospheres and the samples calcined at varying temperature. a) The XRD patterns of Low-Cry-STO with reaction time of 2h and single-crystal STO calcined at 1123 K to 1523 K, suggesting that all the samples are indexed to cubic perovskite SrTiO_3 (JCPDS standard card #35-0734) and less SrCO_3 exists in Low-Cry-STO precursor, b) the enlarged {110} diffraction peak region in (a), the enhanced peak intensity and sharper peak shape of {110} facet characteristic diffraction verified the constantly developed crystallinity with the increasing temperature, and c) The enlarged region ranging from 20 to 40 degrees in (a).

Little amount of SrCO_3 were found to be present in the Low-Cry-STO, which may have been produced by the reaction with CO_2 in air during hydrothermal reaction.

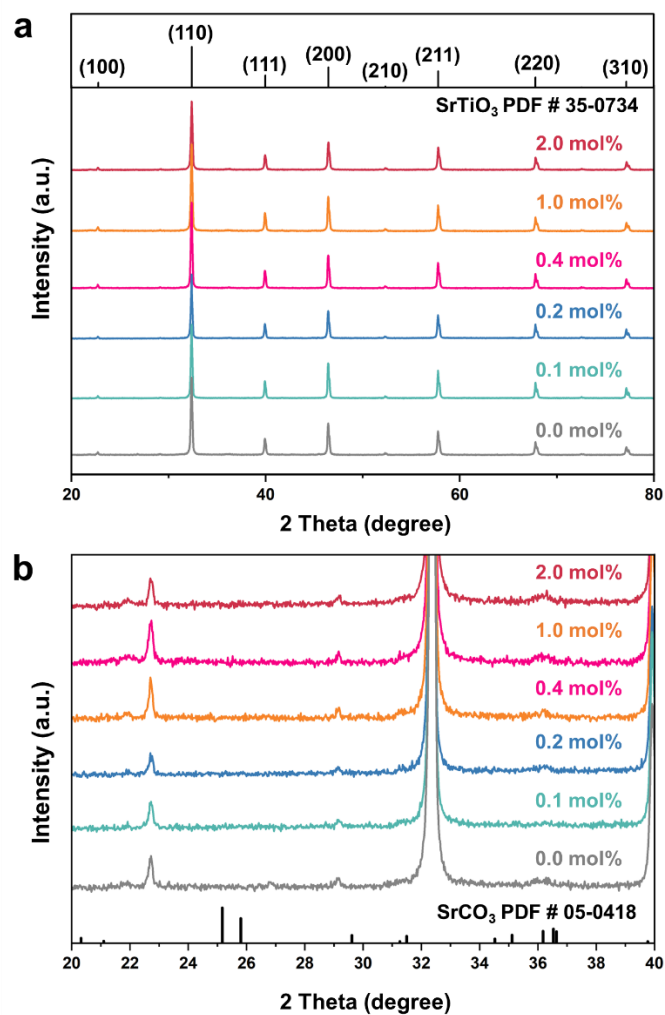


Figure S11 The crystal structure of post-photocatalytic samples with varying amount of Al^{3+} additive. a) The XRD patterns of post-photocatalytic samples calcined at 1423K with various amounts of Al doping, and b) the enlarged region ranging from 20 to 40 degrees in (a). The post-photocatalytic samples for XRD analysis have undergone photocatalytic OWS reaction for 3 hours.

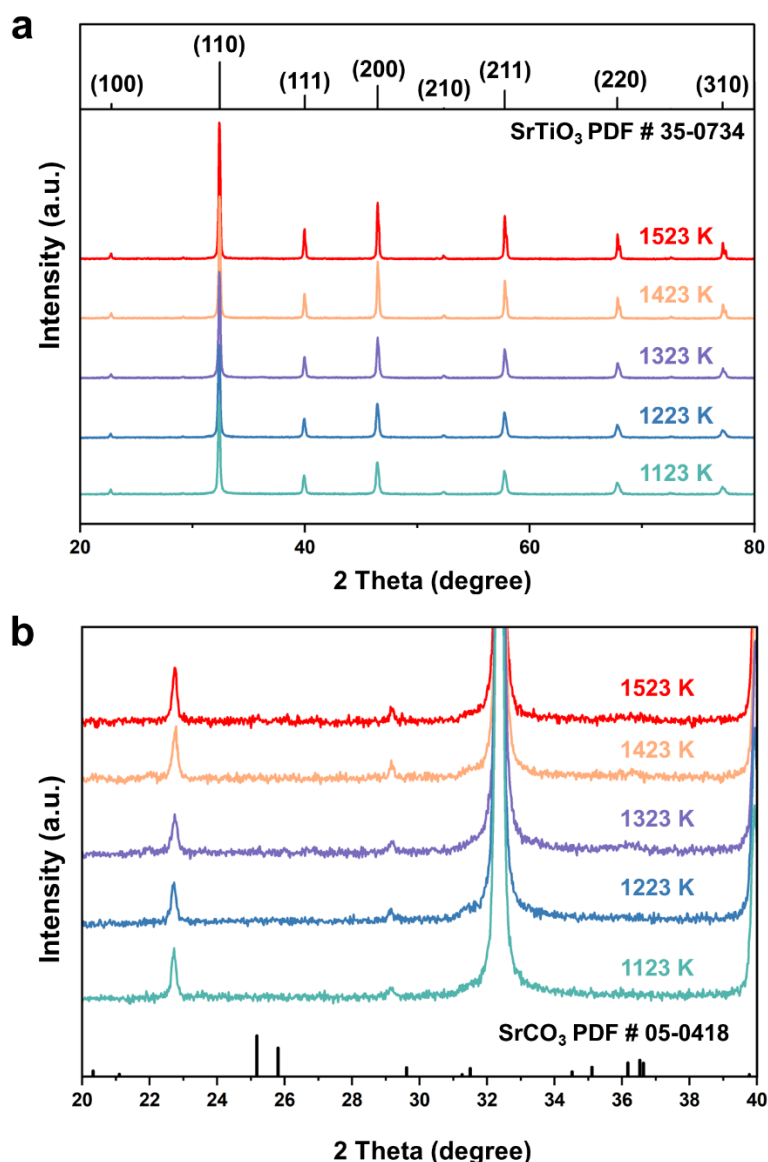


Figure S12 The crystal structure of post-photocatalytic samples. a) The XRD patterns of post-photocatalytic samples, calcined at the temperature ranging from 1123 K to 1523 K, and b) the enlarged region ranging from 20 to 40 degrees in (a). The post-photocatalytic samples for XRD analysis have undergone photocatalytic OWS reaction for 3 hours.

The XRD patterns in **Figure S9** reveal the presence of trace amounts of SrCO₃ in the as-prepared STO single-crystal samples, which undergoes decomposition during photocatalytic OWS reaction. In the samples after a 3-hour reaction, SrCO₃ has disappeared, while **Figure 5e** indicates that the reaction activity remains stable within at least 20 hours. This suggests that the decomposition of SrCO₃ does not affect the photocatalytic activity of STO single-crystal.

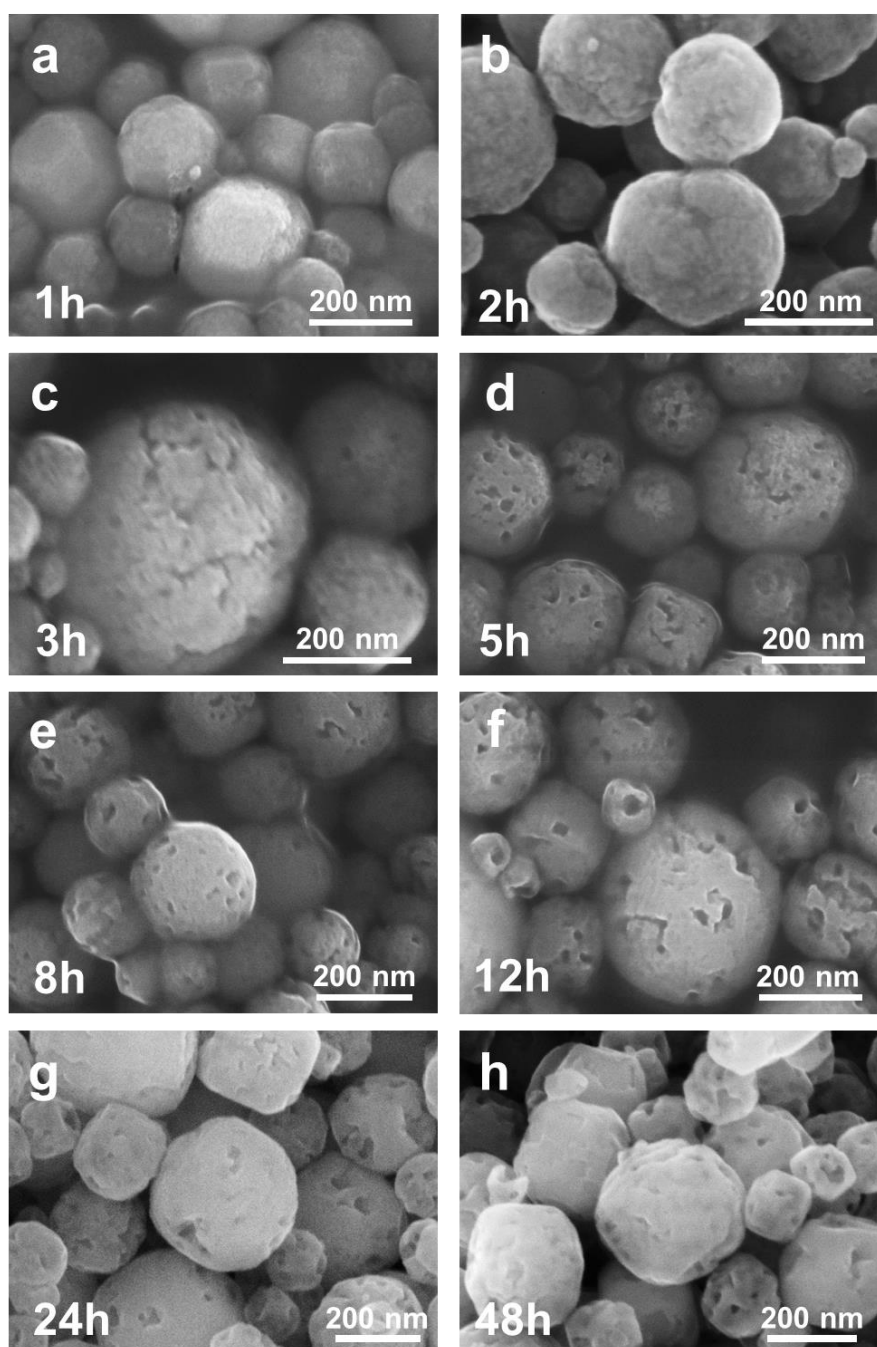


Figure S13 SEM images of various nanosphere STO precursor. SEM images of SrTiO_3 nanosphere precursor synthesized with different reaction time of a) 1h, b) 2h, c) 3h, d) 5h, e) 8h, f) 12h, g) 24h, h) 48h. With the increase of reaction time, the nanoparticle evolved mesoporous structures, consistent with report in the literature.

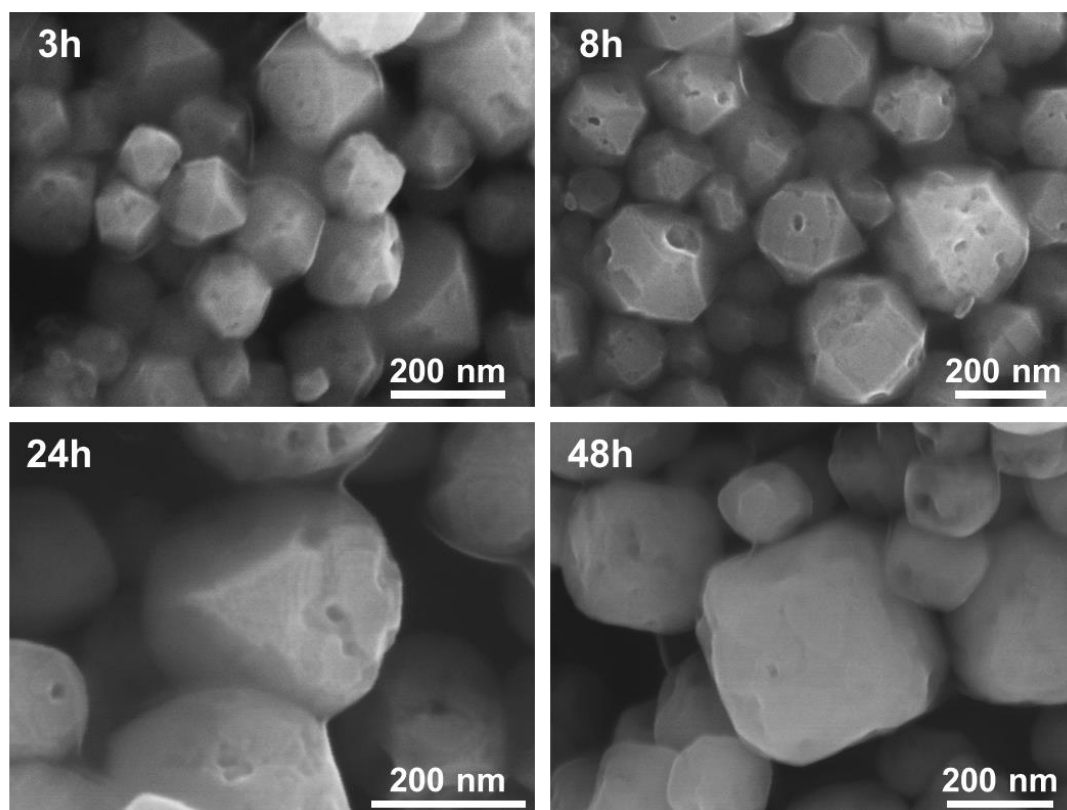


Figure S14 SEM images of cuboctahedron STO calcined at 1173K from various nanosphere precursors. SEM images of cuboctahedron STO prepared by calcination at 1173K from the nanosphere precursors obtained with various reaction time, the time from 3h to 48h marked in the graph indicates the reaction time of the corresponding precursor that shown in **Figure S7**.

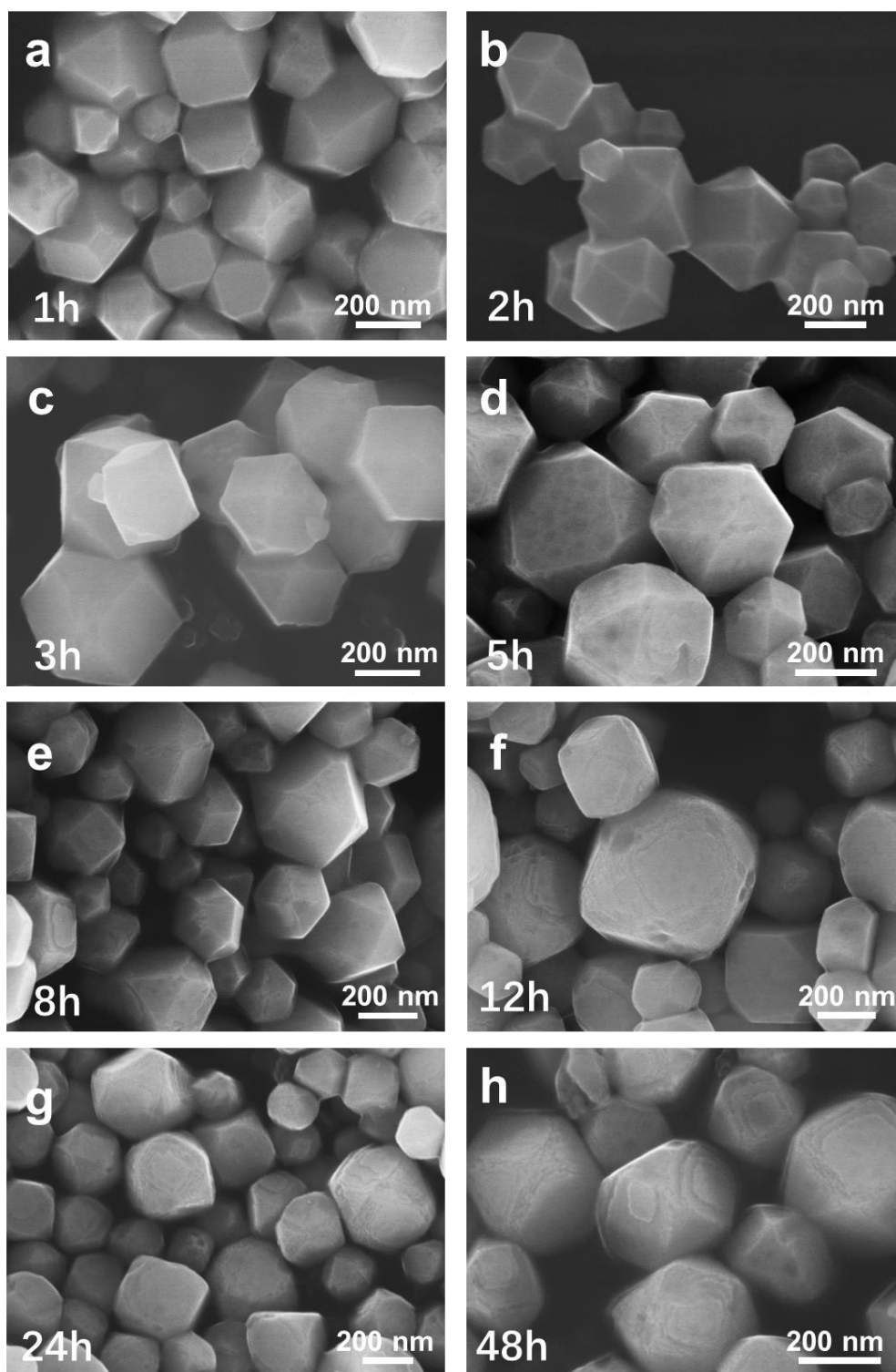


Figure S15 SEM images of cuboctahedron STO calcined at 143K from various nanosphere precursors. SEM images of cuboctahedron STO prepared by calcination at 1423K from the nanosphere precursors obtained with various reaction time, the time from 1h to 48h marked in the graph indicates the reaction time of the corresponding precursor that shown in **Figure S7**.

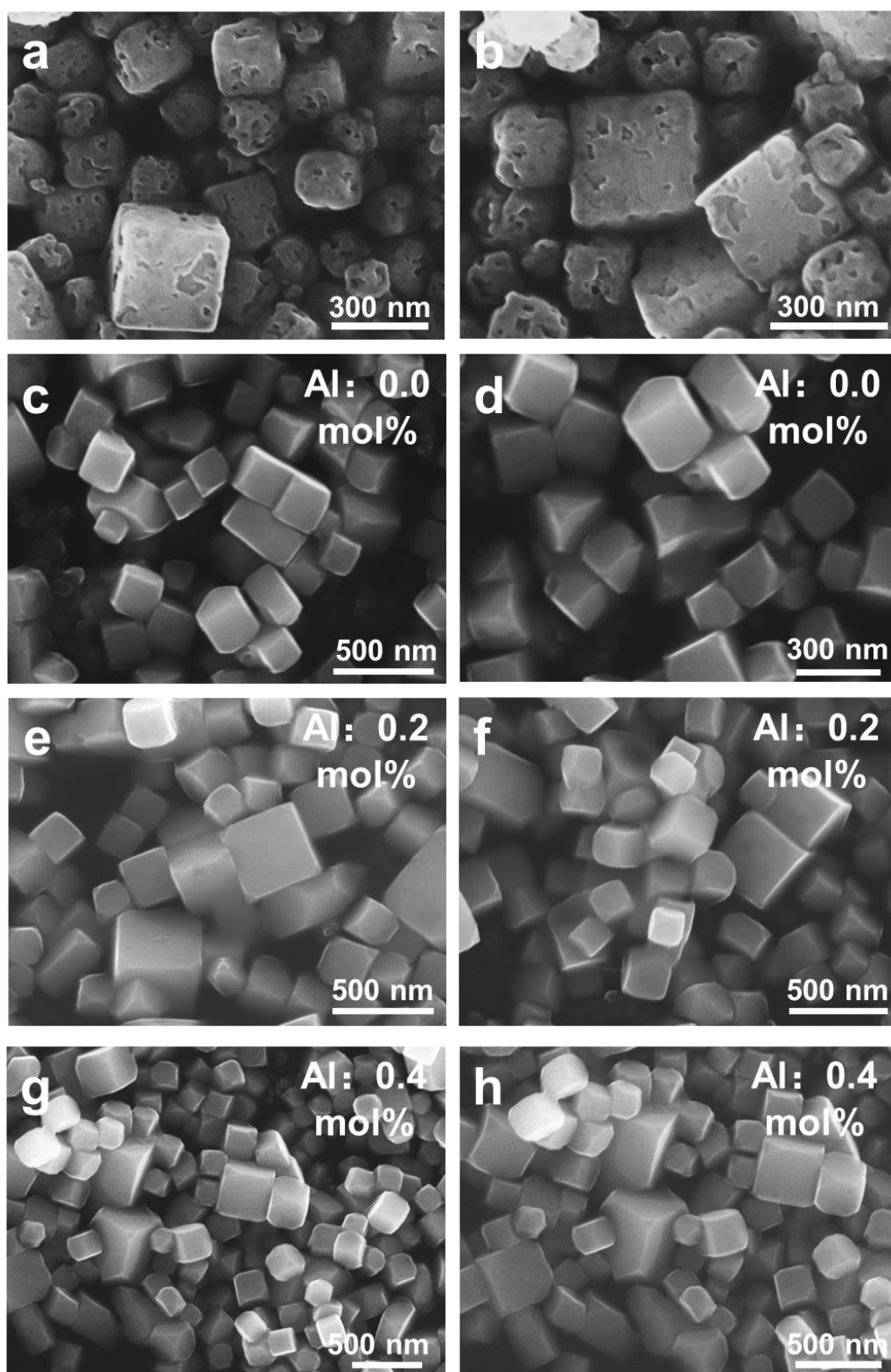


Figure S16 Effect of Al doping on crystal morphology obtained from cubic precursor. SEM images of a-b) mesoporous cubic STO precursor with reaction time of 48 h, c-h) a series of STO single crystals prepared at 1423K for 10h with various amounts of Al doping (0.0~0.4 mol%) using mesoporous cubic STO as precursor.

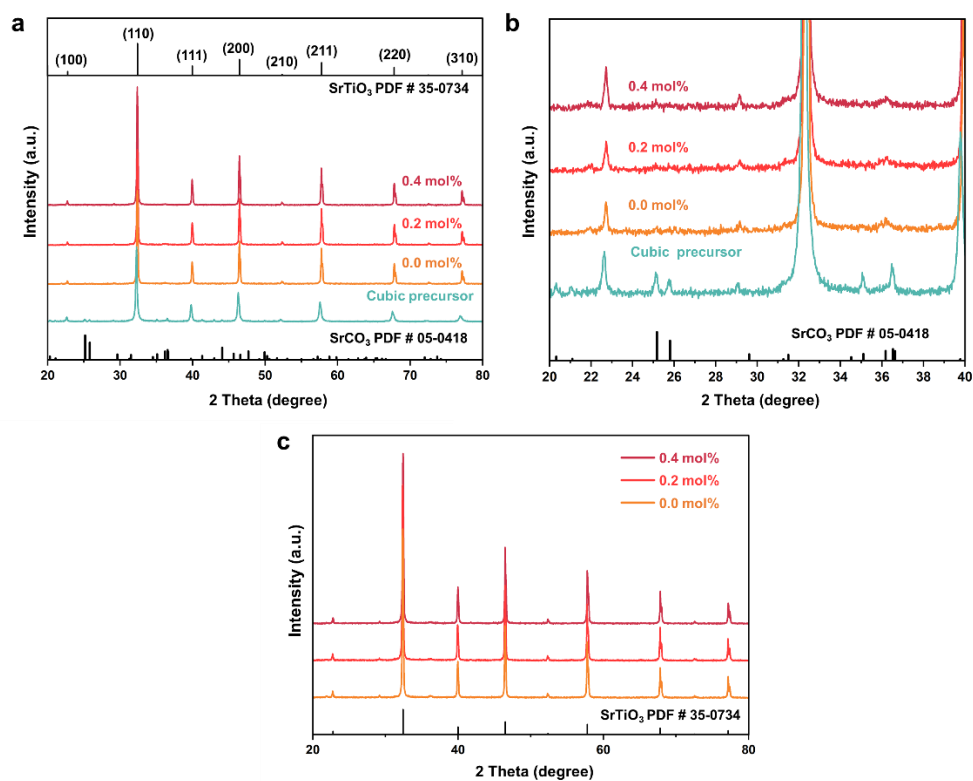


Figure S17 Crystal structure of cubic STO single crystals obtained from cubic precursor. The XRD patterns of a) cubic mesoporous STO precursor and as-prepared STO crystals calcined at 1423K with various amounts of Al doping, b) the enlarged region ranging from 20 to 40 degrees in (a), and c) the corresponding post-photocatalytic samples.

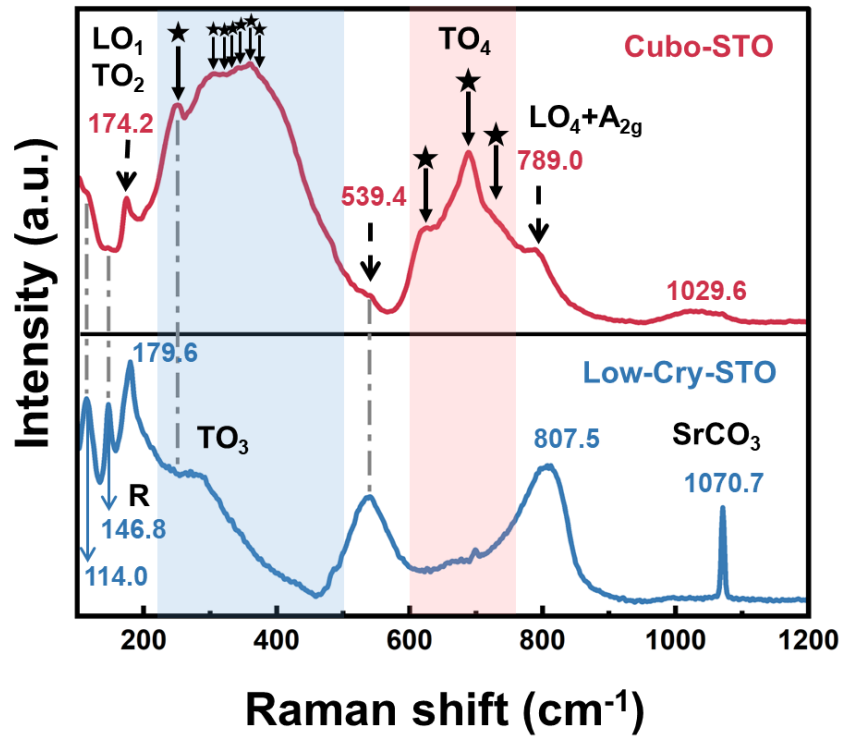


Figure S18 Raman spectra of typical Low-Cry-STO and Cubo-STO calcined at 1423K.

For Cubo-STO, two strong broad Raman peaks of second-order scattering are observed from 220 cm⁻¹ to 500 cm⁻¹ and 590 cm⁻¹ to 760 cm⁻¹, respectively, both of which consist of large number of modes for the typical pattern of STO powder, and the mode observed at 1029.6 cm⁻¹ is also a second-order mode.⁷⁻¹⁰ As for Low-Cry-STO, in contrast, no corresponding vibration modes are observed in these regions, which confirms the major disordered phase. The modes located at 179.6 cm⁻¹, 539.4 cm⁻¹, and 807.5 cm⁻¹ are assigned to the first-order modes, which are symmetrically forbidden in STO with an ideal centrosymmetric cubic structure.⁷ In addition, the sharp signals and obvious blue shift of first-order Raman scattering detected with Low-Cry-STO represent the breaking of symmetry and the existence of polar microregions.^{8,10} Especially, the polar TO₂ and nonpolar TO₃ phonon present a markedly Fano-type asymmetry and a long-range structural distortion.⁹ It is suggested that the local structure strains exist in Low-Cry-STO precursors,¹⁰ and vanish through localized structure rearrangement during recrystallization.

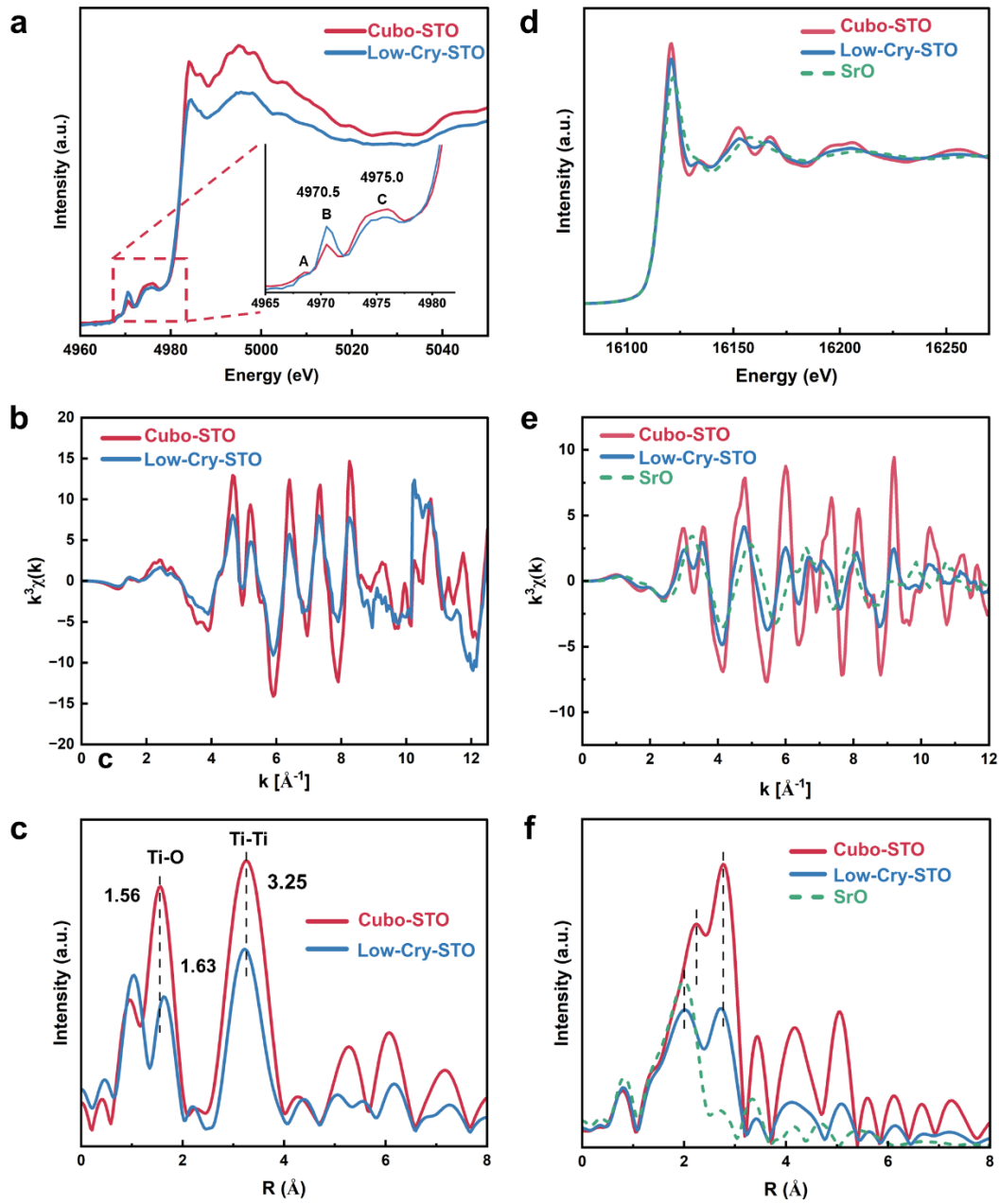


Figure S19 Local structure characterizations of as-prepared Low-Cry-STO and Cubo-STO.

a) Ti K-edge XANES spectra, b) k^3 -weight EXAFS and c) Fourier transformed k^3 -weight EXAFS of Ti K-edge. d) Sr K-edge XANES, e) k^3 -weight EXAFS and f) Fourier-transformed k^3 -weight EXAFS of Sr K-edge.

A typical pattern of three pre-edge peaks (P_A , P_B , P_C) is observed at the Ti K-edge XANES data (**Figure S 19a**), which has been well explored in literatures.¹¹⁻¹⁴ In particular, it is shown that the intensity of the peak P_B is proportional to the displacement of Ti atoms from the center of the TiO_6 octahedra.⁷ The prominent intensity of peak P_B of Low-Cry-STO reveals the loss of centrosymmetry that the Ti ions are present in the severely distorted TiO_6 octahedral structure. The reduced amplitude of Ti K-edge and Sr K-edge X-ray absorption fine structure (EXAFS) spectra of Low-Cry-STO are consequence of disorder effects of EXAFS signals resulted from the amorphous phase.^{11,12} The first peak in the Fourier-transformed EXAFS spectra corresponds to the first

coordination Ti-O shell layer. The second peak is assigned to the Sr-Ti and Ti-O-Ti bonds.¹³ The decrease in the intensity of Ti-O peak and the longer Ti-O bonds in the Low-Cry-STO with respect to the Cubo-STO can be ascribed to the disorder of bond length due to Ti eccentricity displacement, consistent with the XANES data. The evident shift and weakness of peak in the Fourier transform EXAFS signals of Sr K-edge also indicates the loss of local symmetry for Low-Cry-STO.¹¹ Thus, it is believed that the distorted local structure and weak chemical bonding of Low-Cry-STO precursors are responsible for the improved recrystallization kinetics.

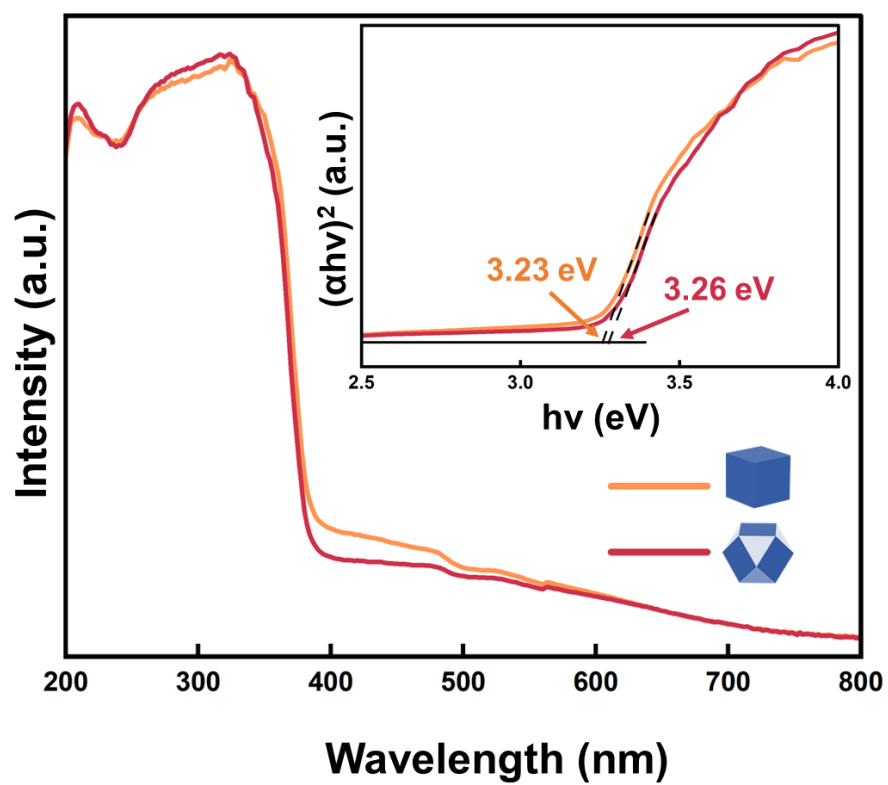


Figure S20 UV-vis DRS and plots of the transformed Kubelka–Munk function for cubic and cuboctahedron STO.

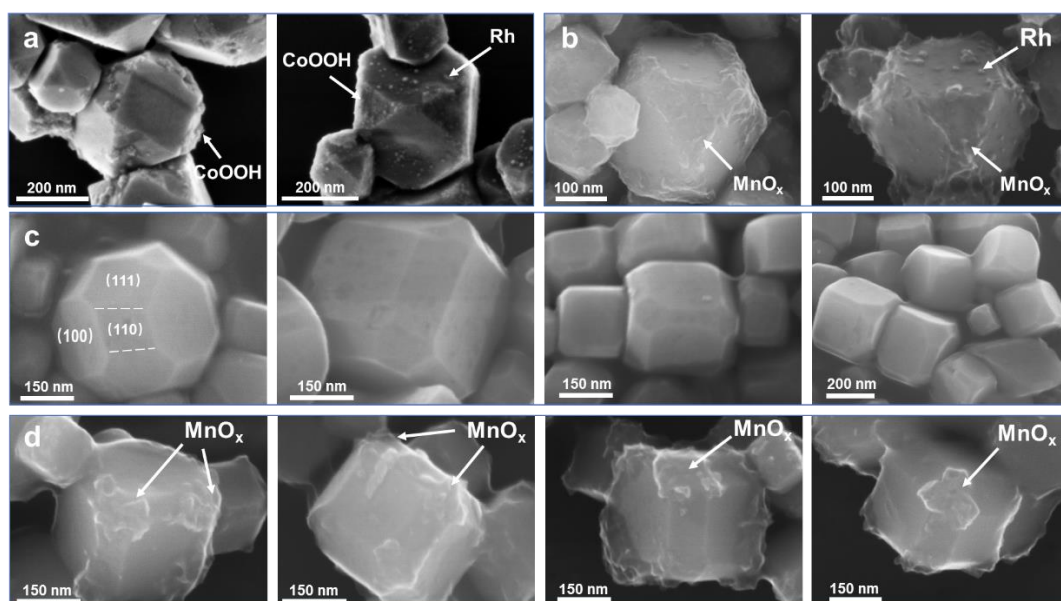


Figure S21 SEM images of single crystals exposed with {100}, {110}, and {111} facets. SEM images of Cubo-STO deposited with a) 0.4 wt% CoOOH and 0.4 wt% Rh + 0.4 wt% CoOOH, b) 0.4 wt% MnO_x and 0.4 wt% Rh + 0.4 wt% MnO_x. SEM images of c) twenty-six-faceted polyhedral STO crystals enclosed with {100}, {110} and {111} facets prepared by calcining at 1223K from nanosphere STO precursor with reaction time of 36h and d) typical crystals loaded with 1.0 wt% MnO₂, indicating the preferential oxidation of Mn²⁺ on {111} facets.

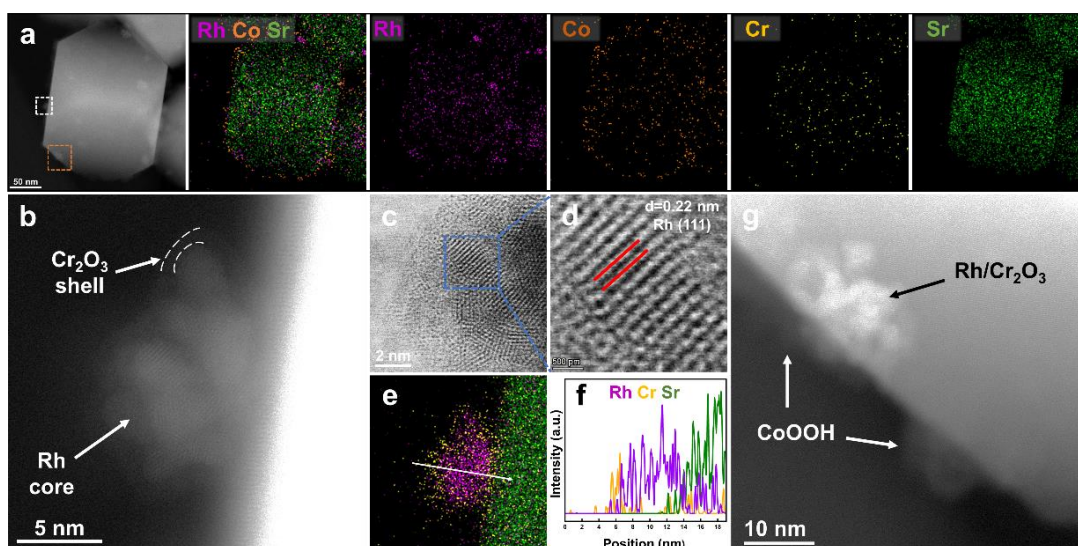


Figure S22 Structure and distribution of cocatalyst. a) Dark field STEM image and corresponding EDS maps of typical STO with partially exposed {111} facet deposited with Rh (0.4wt%)/Cr (0.2 wt%)/CoOOH (0.4wt%). b) HAADF STEM image of white box in (a). c, d) bright field STEM image of Rh/Cr₂O₃ nanoparticle, e) EDS maps of Rh/Cr₂O₃ nanoparticle loaded on {100} facet and f) linear scanning along the white arrow in (e), showing the core-shell structure composed of Rh core along [111] orientation and Cr₂O₃ shell of 2-3 nm thickness. g) HAADF STEM image of orange box in (a), showing the selective deposition of CoOOH on {111} facet and Rh/Cr₂O₃ on {100} facet.

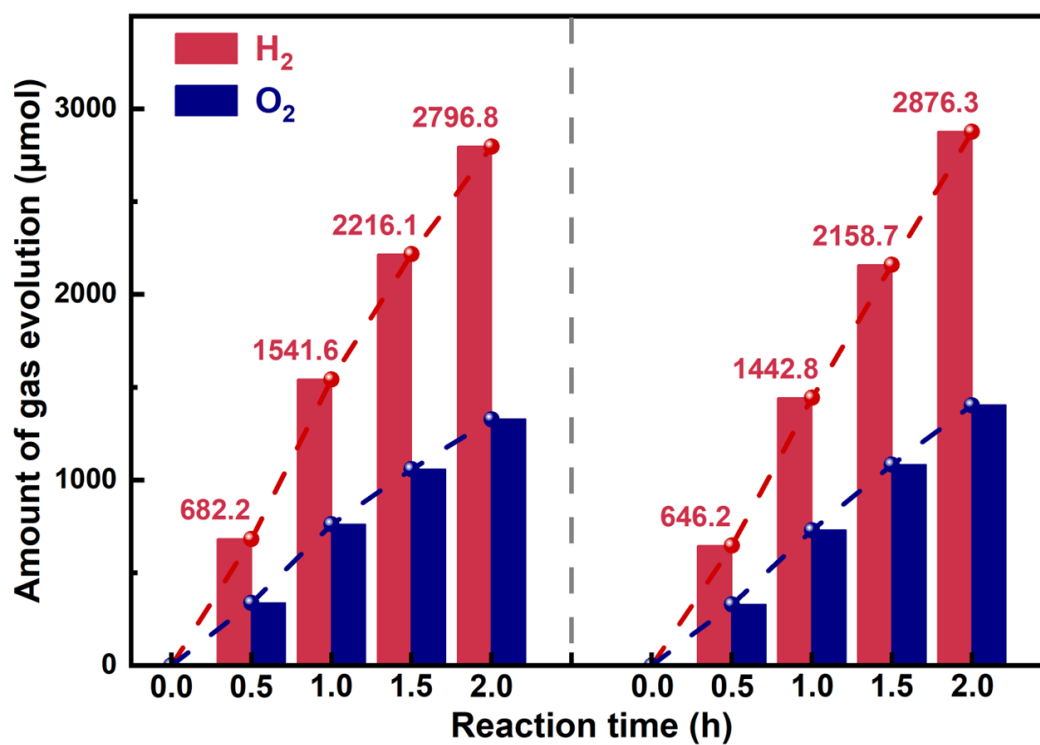


Figure S23 Effect of co-catalyst loading sequence on the OWS activity. Time courses of gas evolution during photocatalytic overall water splitting over cuboctahedron STO modified with Rh (0.2 wt%)/Cr₂O₃ (0.1 wt%)/CoOOH (0.2 wt%) with different procedure of co-catalyst deposition. Left: the photo-deposition with sequence of Rh, Cr, Co. Right: the photo-deposition with sequence of Rh+Co, then Cr.

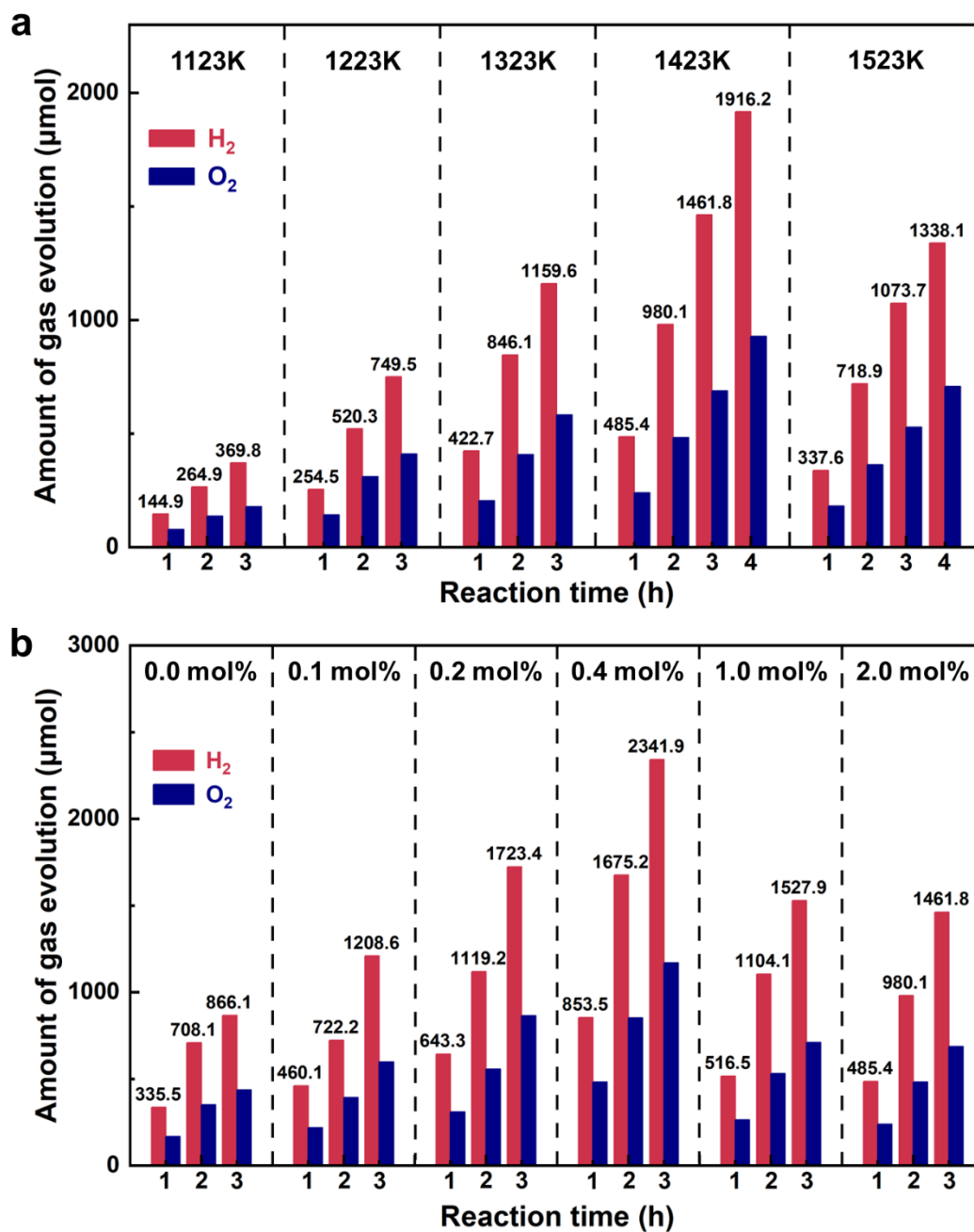


Figure S24 Photocatalytic overall-water-splitting activity. Time courses of gas evolution during overall water splitting over CoOOH/Rh/Cr₂O₃-modified cuboctahedron STO synthesized with a) calcination temperature from 1123K to 1523K and b) amount of Al doping, indicating that the maximum reactivity was achieved by the Cubo-STO calcined at 1423K with 0.4mol% Al doping, corresponding to the most well-defined morphology. Conditions: photocatalyst, 0.1 g; Rh, Cr, Co loading amount on SrTiO₃, 0.1 wt%, 0.05 wt%, 0.1 wt%; ultrapure water, 100 mL; light source, 300 W Xenon lamp (>300 nm).

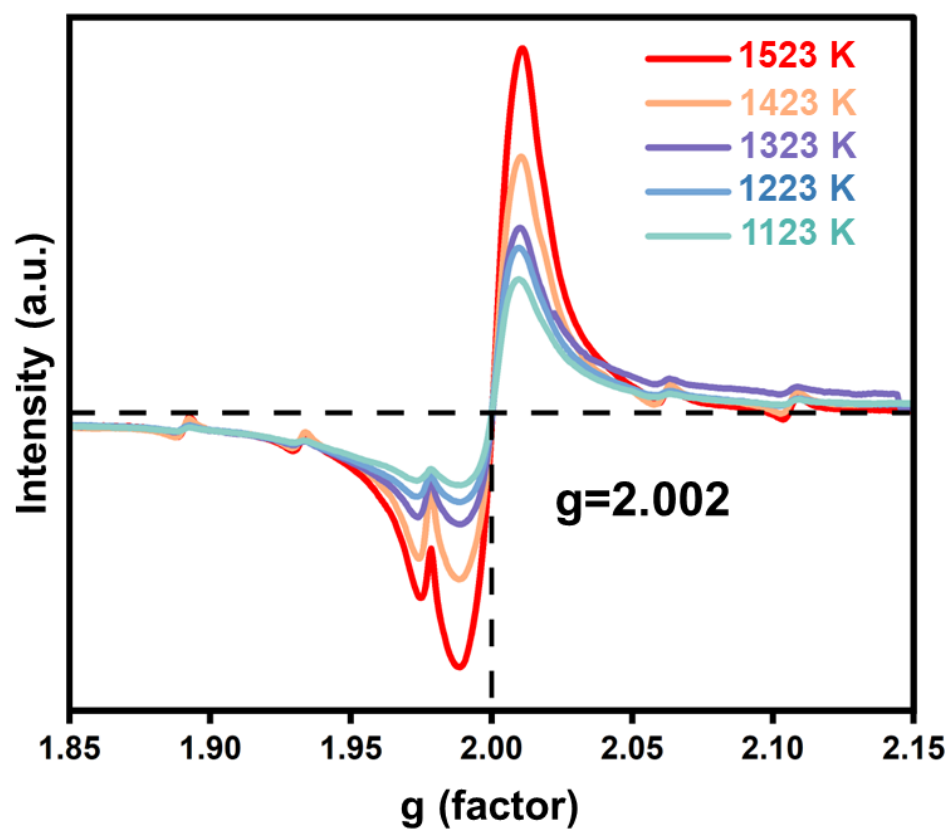


Figure S25 EPR spectra of STO calcined with temperature from 1123 K to 1523 K. The signal at $g=2.002$ should be assigned to oxygen vacancy (V_o).

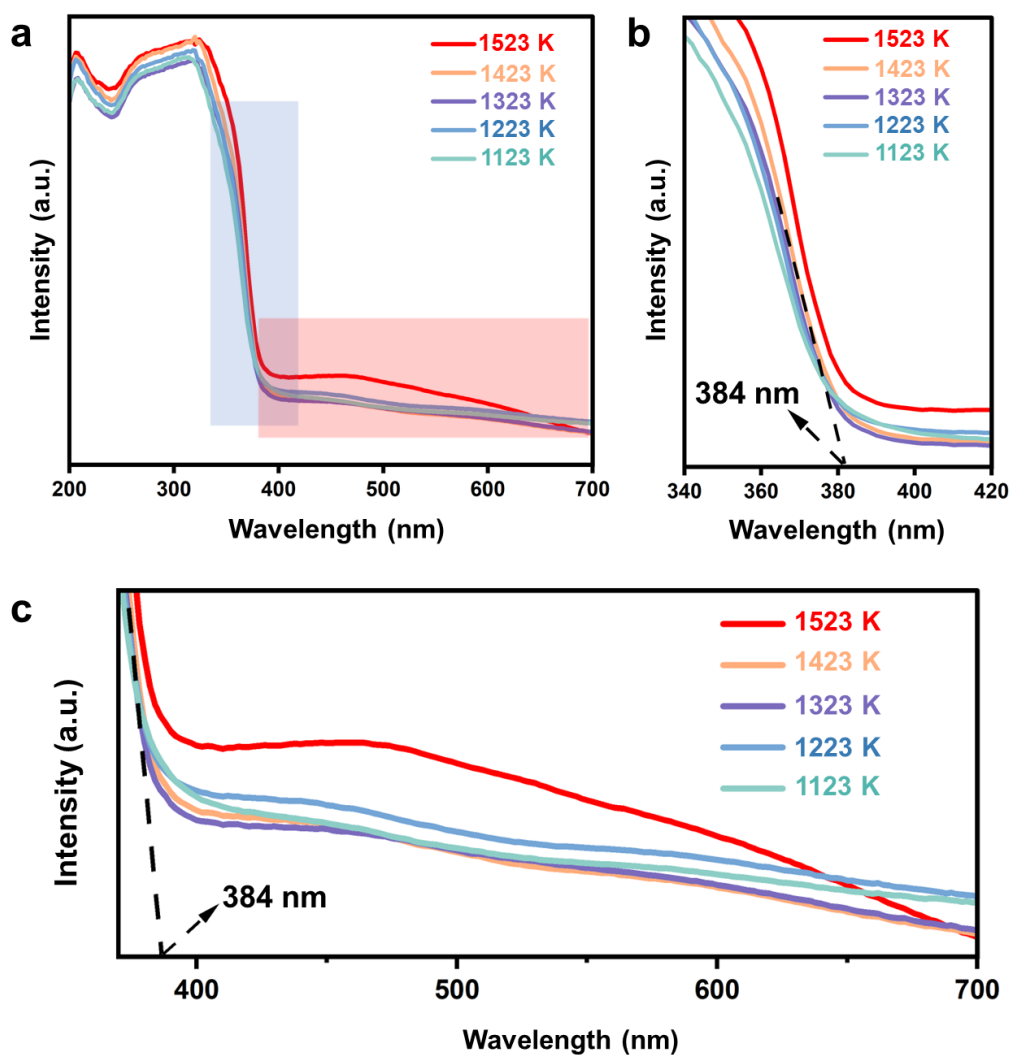


Figure S26 Ultraviolet-visible diffuse reflectance spectroscopy (UV-Vis DRS) of Cubo-STO calcined from 1123K to 1523K. Samples calcined at 1123K-1523K show the same absorption band edge at 384 nm, but the sample obtained at 1423K has the lowest defect absorption above 400 nm.

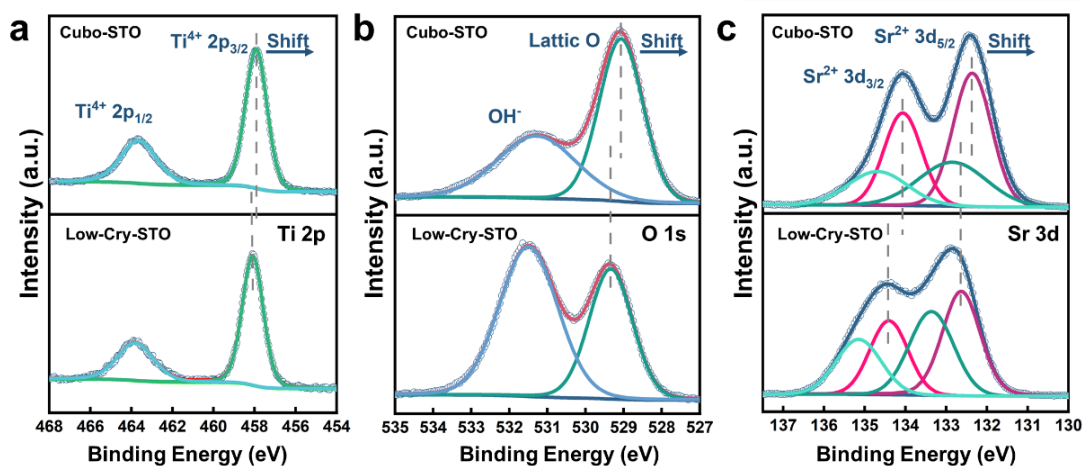


Figure S27 X-ray photoelectron spectroscopy analysis. The XPS spectra for as-prepared Low-Cry-STO and Cubo-STO of a) Ti 2p, b) O 1s and c) Sr 3d.

XPS is used to study the surface chemical composition and state of the as-prepared samples. The Ti 2p spectra (**Figure S10a**) shows a reasonable fit to a single doublet with a 2p 3/2:1/2 peak ratio of c.a. 2:1, consistent with a single Ti^{4+} environment predominating, split by spin orbit coupling into 2p3/2 and 2p1/2 components. The largest component (c.a. 529.5 eV) of O 1s in **Figure S10b** is assigned to O^{2-} (i.e oxide) groups that make up the bulk of SrTiO_3 . The broader intermediate peak (c.a. 531.3 eV) is assigned either to surface OH^- . The Sr 3d spectra of the samples (**Figure S10c**) shows a doublet consistent with peak ratios of c.a. 3:2, consistent with a single Sr^{2+} environment split by spin orbit coupling into 3d 5/2 and 3d 3/2 components.^{15,16}

A consistent shift to lower binding energy is observed in both the Ti, O, and Sr peaks and these changes should be attributed to the shift in the Fermi level position shifting.¹⁶

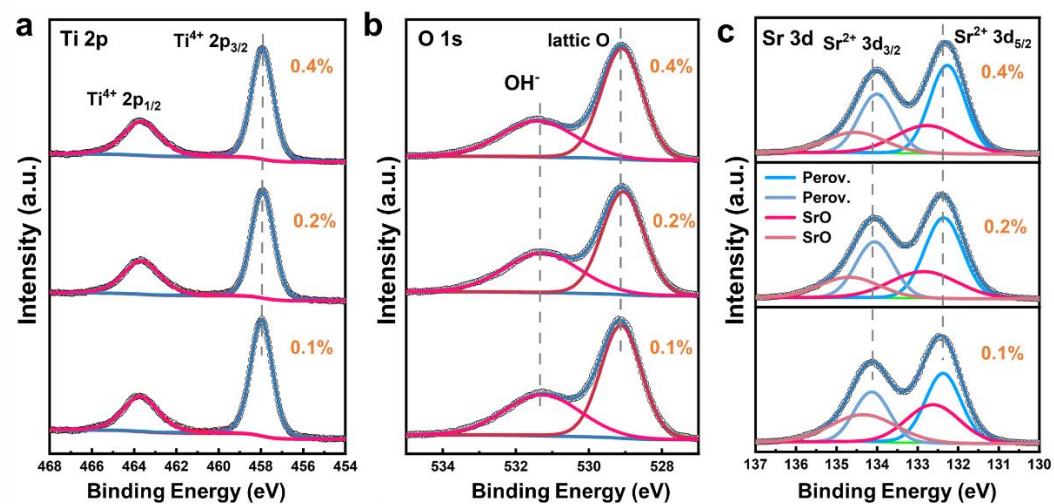


Figure S28 Effect of Al doping on chemical state. XPS spectra of as-prepared STO crystals with various amounts of Al³⁺ addition of a) Ti 2p, b) O 1S and c) Sr 3d.

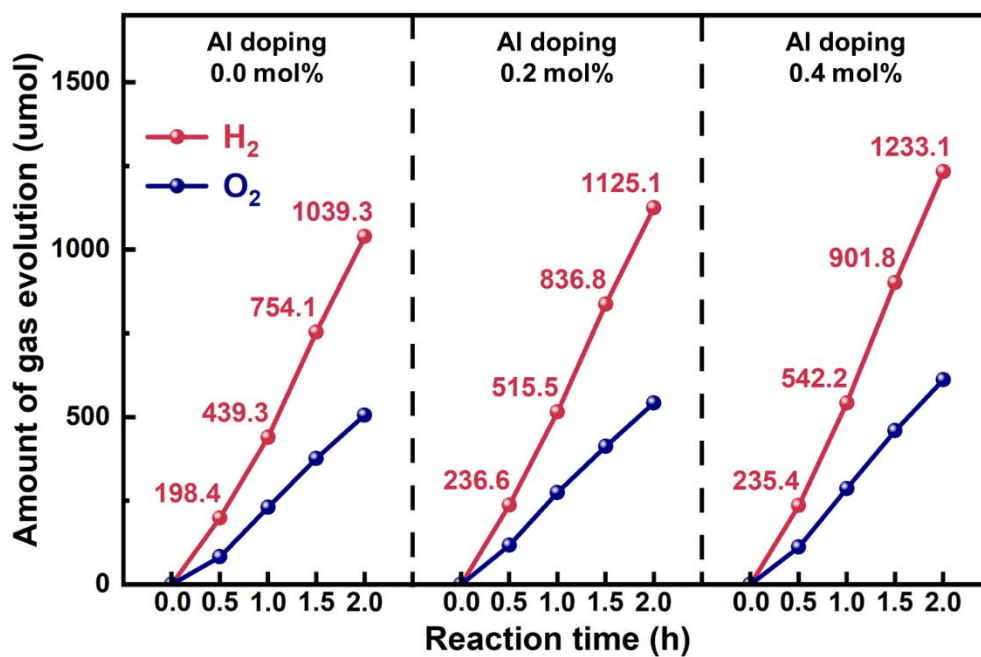


Figure S29 Effect of Al doping on the OWS activity of cubic STO obtained from cubic precursor. Time cause of gas evolution during photocatalytic overall water splitting over cubic STO obtained from cubic mesoporous STO precursor with Al doping varying 0.0-0.4 mol%. Conditions: photocatalyst, 0.1 g; Rh, Cr, Co loading amount on SrTiO₃, 0.1 wt%, 0.05 wt%, 0.1 wt%; ultrapure water, 100 mL; light source, 300 W Xenon lamp (>300 nm).

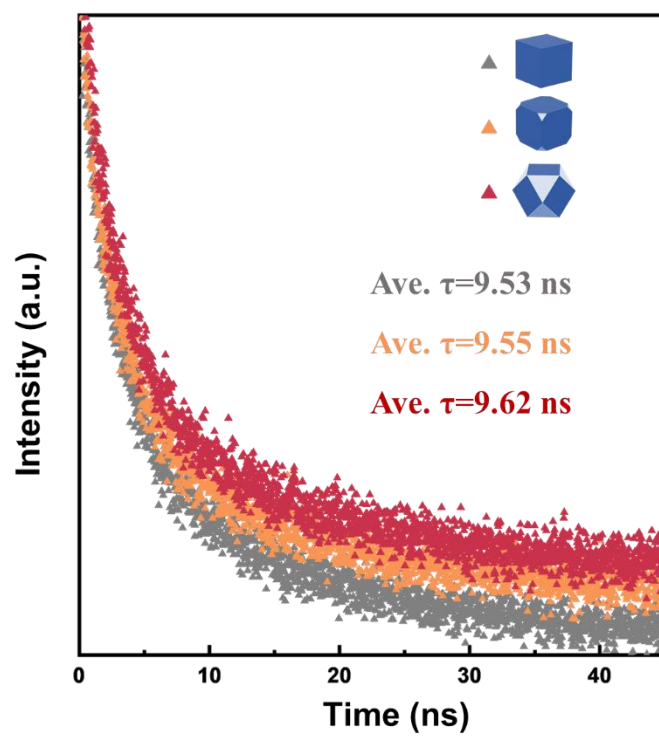


Figure S30 Time-resolved photoluminescence (TR-PL) emission decay spectroscopy. The longest PL emission lifetime illustrates the crucial role of internal electric field between $\{100\}$ and $\{111\}$ facets in inhibiting the recombination of photoexcited carriers.

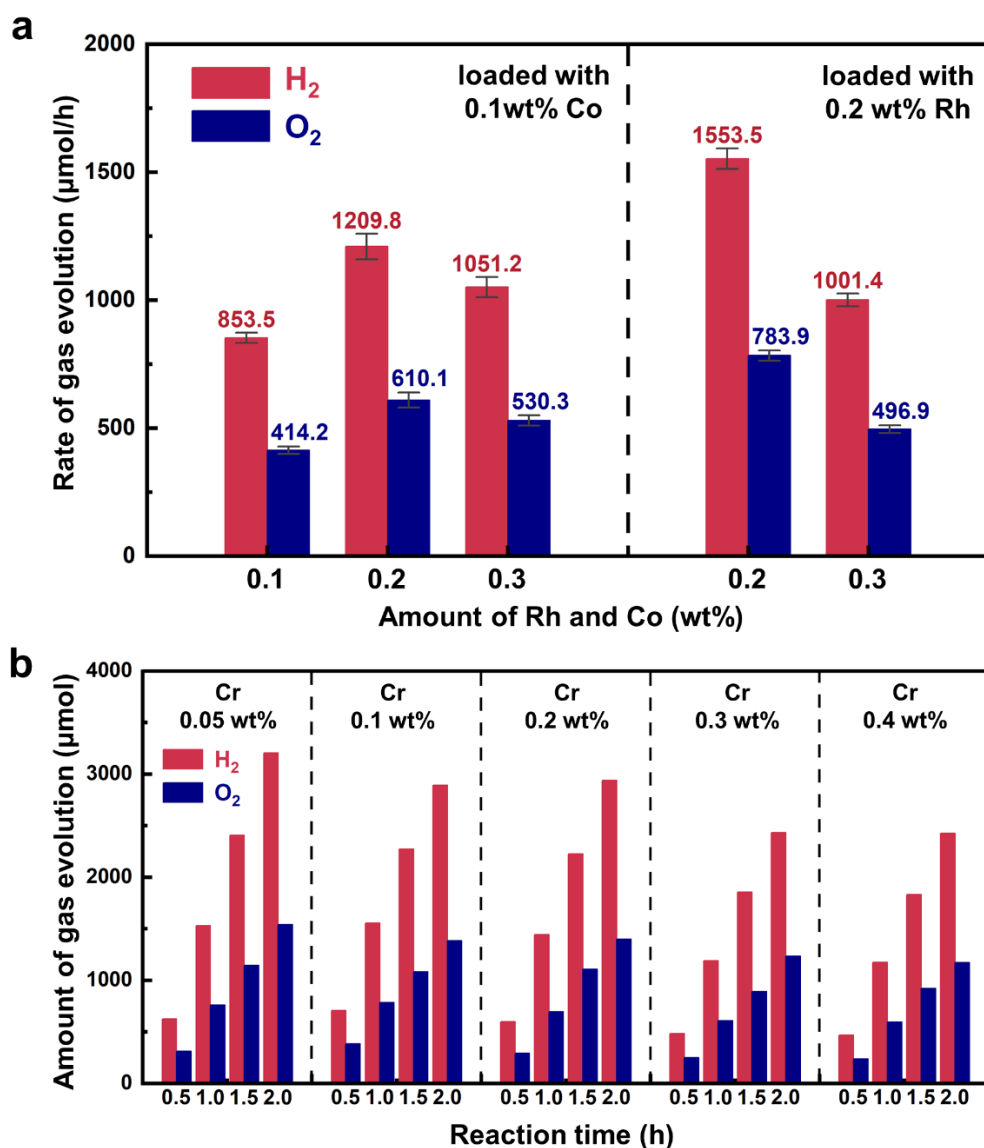


Figure S31 Effect of co-catalyst loading mass on the OWS activity. a) Dependence of gas evolution rates on Cr₂O₃/Rh/CoOOH modified Cubo-STO powder upon the amount of Rh (Left) and CoOOH (Right). The loading amounts of Co in left and Rh in right were kept 0.1wt% and 0.2wt%, respectively; the amount of Cr was always half of Rh. b) Dependence of photocatalytic OWS gas evolution on Cr₂O₃/Rh/CoOOH modified Cubo-STO powder upon the amount of Cr (ranging from 0.05-0.4 wt%). The loading amounts of both Rh and Co were always kept as 0.2wt%. The amounts of OWS gas evolution remain at a consistent level within the Cr loading range of 0.05-0.2 wt% and a gradual decline in performance was observed only when the Cr loading exceeds 0.3 wt%. Thus, an optimized loading mass of Rh (0.2 wt%)/Cr₂O₃ (0.1 wt%)/CoOOH (0.2 wt%) was used.

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